

From Terrestrial Laser Scans to Queryable Robot Knowledge: A VLM-Verified Framework for Incremental 3D Semantic Modeling

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Abstract: Indoor service and inspection robots need environment knowledge that is queryable, reliable, and maintainable as the site changes, yet open-vocabulary 3D pipelines propagate unverified labels downstream and build-once scene models offer no maintenance path. We present TLS-SMF, a semantic modeling framework that converts registered terrestrial laser scans into a confidence-aware semantic knowledge graph and keeps it current across repeated scans. TLS-SMF combines (i) a deterministic geometric backbone whose re-runs are byte-identical on the same host, (ii) multi-view vision–language-model verification with automatic escalation that retains unresolved objects as explicitly unverified, and (iii) identity-preserving incremental updates that apply confidence-gated node-level upserts to mint-once node identities. Across two operational configurations of one industrial room, escalation lifts verification accuracy from 77.8% to 88.9% and from 78.1% to 84.4%, with 94.4–100% unanimous-vote precision against confirmed owner ground truth; on the showroom benchmark, structurally gated querying answers 83.3% of questions versus 52.1% for an unverified Stage-A LLM baseline while blocking hallucination traps; in a controlled same-quality combined-edit re-scan, selective re-verification reduces VLM cost to 3.1% of a full rebuild; and a real three-epoch study detects 2/2 controlled physical changes with zero observed identity switches on strong reference correspondences, while re-segmentation still produces false-new / false-absent cases. A cross-floor cafeteria stress scene bounds domain transfer, and an owner-in-the-loop revision path keeps the single semantic store corrigible for robot-facing use; the same store drives a physical mobile manipulator, which completes three console-issued semantic missions with a mean arrival error of 0.30 m while phantom goals are refused with zero motion.

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1. Introduction

Mobile robots deployed in industrial interiors increasingly need more than a map: they need environment *knowledge*—which objects exist, where they are, what they afford—that can be queried in task language, trusted enough to act on, and kept current as the site changes. Terrestrial laser scanning (TLS) is an attractive source for such knowledge: survey-grade scanners deliver millimeter-accurate, globally registered geometry of an entire site in minutes, and are already routine in construction and facility management. Yet the path from a raw scan to knowledge a robot can safely use is broken in three places.

First, *semantic labels do not survive contact with industrial scenes*. Open-vocabulary 3D classifiers trained on web-scale or indoor-furniture distributions degrade sharply on industrial content: in our experiments a state-of-the-art point–text embedding model

misclassifies most equipment, and—counterintuitively—restricting its vocabulary to the classes actually present makes accuracy *worse* (66.7% $\rightarrow$ 33.3%, Sec. 5.8), indicating an embedding-level domain gap rather than a class-list problem. Pipelines that propagate such labels downstream hand the robot confident fictions.

Second, *ungrounded language interfaces hallucinate*. Letting an LLM answer questions about a scene from its own priors, or from an unvalidated label dump, produces fluent but unsupported claims—20.8% of atomic factual claims in our LLM-only benchmark condition (Sec. 5.5). For a robot, a hallucinated fire extinguisher is not a harmless chat artifact.

Third, *scene models are built once and abandoned*. Most semantic mapping pipelines produce a single-epoch artifact with no principled update path: a re-scan either rebuilds everything—losing object identities that downstream planners and logs refer to—or is not integrated at all. Real sites drift continuously; in our own test room, roughly one third of object geometry changed within two weeks (Sec. 5.6).

This paper presents **TLS-SMF** (TLS-driven Semantic Modeling Framework), which addresses all three failures with one design principle: *separate what can be made deterministic from what must remain probabilistic, and control the probabilistic part with explicit confidence machinery*. Concretely, we contribute:

1. **A deterministic geometric and explicit-modeling backbone.** From registered TLS input, seeded-RANSAC structure removal, fixed-parameter clustering with a conservative giant-split and a structure-remnant filter, and PCA-based explicit modeling run fully automatically; same-host re-runs are byte-identical. Determinism is not an implementation nicety: it is what makes verification results carryable, diffs meaningful, and, under reproducible acquisition, allows verification cost to track detected content change.
2. **Confidence-aware multi-view VLM verification with automatic escalation.** Each object is classified per-view under a forced answer schema; votes are fused confidence-weighted; split votes escalate automatically to a wider view set; and objects that still fail to converge are retained as explicitly *unverified*—never silently dropped and never resolved by a human in the loop.
3. **Identity-preserving incremental knowledge-graph update.** Node IDs are minted once and carried by two-pass matching (spatial, then type-constrained move detection); re-scans apply node-level unchanged/updated/moved/inserted/absent upserts with per-revision history, and unchanged objects re-verify for free by content-hash carry-over.
4. **Corrective decomposition and condition-varied detection.** Diffuseness and wall-compound gates turn “too messy to be an object” from a discard decision into an automatic re-splitting trigger (wall-band stripping, support-plane and panel-bank cuts), and a structure-condition ensemble re-detects under varied wall/ceiling removal, recovering wall-flush objects any single condition amputates while exposing label instability as a per-object attribute.
5. **Owner-in-the-loop revisions over a single semantic store.** The room occupant’s refutations, restorations, and corrections of labels, geometry, and attributes enter as highest-trust but time-variant evidence through the same revision mechanism as re-scans; the management console, the mission mediator, and an Isaac Sim twin are all derived from the one revisioned graph, so a correction propagates to every consumer from a single write; the same mediator drives the physical AMMR (Sec. 6.3).

We are explicit about what is new versus assembled. The base algorithms are deliberately standard and claimed as integration: RANSAC, DBSCAN, and PCA modeling (contribution 1’s backbone), open-vocabulary 3D classification, per-view VLM querying, and the TOSM object/place/robot schema follow prior work. The new mechanisms are the control structures wrapped around them: the confidence-weighted escalation tiers with structural query gating (contributions 2 and the gate of Sec. 4.8), mint-once identity with history-preserving upserts (contribution 3), the trigger-defined corrective layers (contribution 4, Sec. 4.7), and owner feedback as first-class, time-variant evidence (contribution 5).

The additions are quantified against their bases where a base exists: the corrective layers lift automated recall over the raw backbone from 28% to 85% cumulatively (Sec. 5.7); structural gating changes trap hallucination from 0/4 blocked (same graph, ungated) to 3–4/4 (Sec. 5.5); escalation adds +11.1/+6.3 accuracy points over single-pass fusion (Sec. 5.3); and, for context rather than like-for-like comparison, the TOSM lineage reports 78–81% recognition accuracy under a different evaluation protocol, whereas the frozen re-run reaches 85% automated recall with, additionally, per-fact confidence tiers and revision provenance that the lineage does not maintain (Sec. 5.7).

We validate the framework on two operational configurations of one industrial room (an exhibition showroom and a robot test hall), each with owner-confirmed ground truth—plus a cross-floor cafeteria stress scene that bounds domain transfer (Sec. 5.8)—through five benchmark families: segmentation repeatability and sensitivity, verification accuracy and escalation cost, a four-condition knowledge-grounded query benchmark with hallucination traps, synthetic update scenarios, and a real three-epoch study of one room spanning a manual scan, an autonomous exploration scan, and a controlled-change re-scan at 99.6% coverage—plus a real-robot check in which the physical AMMR executes console-issued semantic missions from the same graph (three completed drives, mean arrival error 0.30 m; phantom goals refused; Sec. 6.3). Beyond the headline gains—escalation +11.1/+6.3 points, gated querying +31 points over LLM-only with 3 of 4 hallucination traps blocked, re-verification at 3.1% of rebuild cost on a controlled same-quality re-scan—the study surfaces failure modes we believe are as valuable as the successes: phantom objects from ceiling soffits, unanimous-but-wrong verifications, clutter absorption of small low-reflectance objects, and a coverage paradox in which a *better* scan over-merges density-based clustering. Each is analyzed and either fixed by a method component or reported as a bounded limitation.

The framework extends our ICTC 2026 multi-view VLM classification study with escalation, structural gating, identity matching, incremental update, and the multi-epoch evaluation; the knowledge-graph lineage follows our ISIS 2025 work. Both are cited and the deltas are stated explicitly in Sec. 2.

2. Related Work

2.1. 3D Semantic Mapping and Scene Graphs

Object-level semantic mapping has progressed from object-oriented SLAM and dense semantic fusion [1,2] to metric–semantic pipelines such as Kimera [3] and real-time hierarchical scene graphs [4]. 3D scene graphs organize metric reconstructions into object-level nodes with semantic attributes and inter-object relations [5], and recent open-vocabulary variants attach language-queryable features to nodes and edges [6,7], enabling LLM-driven mobile manipulation over the graph [8]; on the knowledge side, semantic maps have been coupled to task planning since Galindo et al. [9], grounded in spatial databases [10], and served from ontology-backed infrastructures such as KnowRob [11] (see [12] for an application-oriented survey of ontology-based robot knowledge). These systems target flexible *construction*; verification of what enters the graph and maintenance after construction receive less attention. Within the TOSM (triplet ontological semantic model) lineage that this work extends, the original semantic modeling framework [13] defined the explicit/implicit/symbolic triplet; TOSMNav [14] scaled it to a navigation framework in which an OWL/SWRL knowledge base feeds hierarchical PDDL planning—per-robot on-demand databases, kinematics-aware traversability rules that check a robot’s footprint against a doorway’s entrance size, and re-planning with map maintenance when execution fails—validated across multi-floor and outdoor sites with four robot types; and its successor DK-SMF [15] automates object *and place* modeling from RGB-D exploration, combining zero-shot detection and VQA with domain-knowledge filtering and LLM-derived place semantics over SLIC-segmented regions. The navigation side of the lineage *assumes* a trustworthy semantic database; TLS-SMF supplies that database—a verified, incrementally maintained environment model—and its mission mediator (Sec. 6.1) is a

Table 1. Positioning against DK-SMF [15]. The table is the authors’ own tabulation: the DK-SMF column paraphrases that paper’s published description and reproduces none of its tables or figures.

Axis	DK-SMF	This work
Sensing	RGB-D exploration	registered TLS (BLK360)
Label trust	domain-knowledge filtering	multi-view verification + escalation
Lifecycle	build-once	mint-once IDs, incremental upsert
Place semantics	SLIC + LLM over detections	SLIC + ring codes over verified objects

deliberately lightweight instance of the consumption pattern TOSMNav develops in full. Our framework—TLS-SMF, named within the same lineage’s convention of prefixing the differentiating axis—adopts the same target representation (TOSM object/place layers, VDA5050-style export [16]) but differs on four axes, summarized in Table 1: a survey-grade TLS backbone instead of RGB-D streams; multi-view VLM *verification with escalation* instead of domain-knowledge hallucination suppression; identity-preserving *incremental* upserts instead of build-once modeling; and a place layer derived from *verified objects only*, so the confidence gate propagates across layers.

2.2. Open-Vocabulary 3D Recognition and Its Limits

Point-text embedding models such as Uni3D [17] and PointCLIP V2 [18] transfer web-scale vision-language supervision [19] to point clouds, and scene-level systems attach open-vocabulary features to 3D points and instances [20,21]; all perform well on curated indoor benchmarks. On industrial TLS content, however, we observe the transfer breaking down: PointCLIP V2 collapses to a single class, and Uni3D’s accuracy *drops* as the vocabulary is narrowed toward the true class list (Sec. 5.8)—the embedding cannot separate the domain’s classes regardless of prompt curation. We therefore treat open-vocabulary output as a provisional prior to be verified, not a label source.

2.3. Instance Segmentation: Learned versus Geometric

Learned 3D instance segmentation built on point-set backbones [22] (Mask3D [23], SPFormer [24], PointTransformer V3 [25]) and promptable point segmentation (PointSAM [26]) dominate benchmark suites but inherit their training distributions; our transfer attempts to industrial TLS content failed without target-domain fine-tuning, for which no labels exist in our setting. We instead use a deliberately simple geometric decomposition—seeded plane removal [27,28] plus fixed-parameter density clustering [29]—and position it not as a novel algorithm but as a *reproducible decomposition* whose determinism underwrites verification carry-over and incremental diffing (Sec. 4.2). PointSAM is evaluated only as an optional diagnostic refinement of flagged merged clusters under a pre-specified adoption rule (Sec. 5.2); it is not part of the frozen TLS-SMF pipeline.

2.4. Multi-View VLM Verification

Using VLMs to label rendered 3D content is increasingly common; our prior work [30] established multi-view forced-schema VLM classification of TLS object clusters with confidence-weighted late fusion. This paper extends that mechanism into a verification system: automatic escalation of split votes to a wider view set, explicit retention of unresolved objects as *unverified*, tier-precision analysis (including unanimous-but-wrong failure cases), and integration of the resulting status into gating, update, and place layers. The comparison in Sec. 5.3 quantifies the delta over the 4-view baseline.

2.5. Knowledge-Grounded Querying and Hallucination Mitigation

Grounding LLM answers in retrieved structure reduces but does not eliminate unsupported claims [31,32], and instruction-based guardrails are known to be brittle. Our knowledge-graph lineage [33] observed hallucinated environment facts when querying ungrounded LLMs about mapped spaces. We contribute a structural mitigation: rather

Table 2. Capability comparison of environment representations. Each “this work” cell cites the section that substantiates it experimentally; the 2D-map and 3D-scan columns state definitional properties of those representations [39,41] rather than measured results.

	2D map	3D scan	3D semantic KG (this work)
Metric navigation	✓	✓	✓ (derived; Sec. 6.1)
Object geometry	–	✓	✓ (explicit models; Sec. 4.3)
Task-language query	–	–	✓ (gated; Sec. 5.5)
Per-fact confidence	–	–	✓ (tiers; Sec. 5.3)
Change maintenance	re-map	re-scan	node-level upsert (Sec. 5.6)
Place semantics	–	–	✓ (ring codes; Sec. 4.9)

than instructing the model to respect verification status, the query-time view demotes unverified types to *unknown* and quarantines the failed label in a schema-documented field—an intervention in the data contract whose effect we isolate in an ablation (traps 1/4 → 4/4, Sec. 5.8).

2.6. Incremental Semantic-Map Maintenance

Change detection in point clouds and lifelong mapping address geometric currency [34], and object-permanence reasoning tracks instances through rearrangement in embodied settings [35]. Recent online open-vocabulary systems bring incrementality to instance-semantic maps—class-agnostic instance maps labeled zero-shot from selected views [36], online panoptic Gaussian-splatting maps [37], and functional scene graphs that attach interaction semantics to open-vocabulary elements [38]—but what they maintain is a geometric or embedding-level instance representation, not verified semantic evidence. In the TOSM lineage, TOSMNav [14] maintains its map online—updating object positions from detections as robots work, and pruning connectivity when a passage fails—which complements rather than covers our concern. What is rare is maintenance of a *semantic knowledge graph* whose nodes carry verified labels, revision provenance, and stable robot-facing identities across scan epochs: scan-once-build-once remains the default in TLS-derived modeling, whose dominant line reconstructs as-built BIM or parametric building models from static scans [39,40]. Our mint-once identity policy with two-pass matching and confidence-gated node-level upserts (Secs. 4.5–4.6) targets this gap, and Sec. 5.6 evaluates it on synthetic scenarios and a real three-epoch sequence with a controlled-change protocol.

2.7. Map Representations for Robot Tasking

Table 2 contrasts the representations a deployment can choose from. A 2D occupancy map supports navigation but neither queryability nor semantics; a raw 3D scan adds geometry but no task language; a verified semantic knowledge graph adds queryability, per-fact confidence, and—with this work—a maintenance path. The table is analytical; its query and update columns are substantiated by Secs. 5.5 and 5.6.

3. System Overview

Figure 2 shows the framework as a staged pipeline with four loops. A registered TLS scan enters the *deterministic geometric backbone* (blue in the figure): preprocessing with room-scoped structure removal, object decomposition with remnant filtering, automatic *re-split gates* that turn diffuse webs and wall-flush compounds into decomposition triggers rather than discards, and explicit modeling. A parallel *structure-condition ensemble* re-detects under varied removal conditions, contributing wall-flush candidates that any single condition amputates and a per-object label-stability signal. The backbone’s output feeds the *probabilistic semantic stages* (orange): multi-view VLM verification with a height-level physical prior and an internal escalation loop, then confidence-gated integration into the knowledge graph. The graph is the *single semantic store*: the management console, the mission mediator, and the Isaac Sim twin are all derived from the same revisioned file. Two further loops close over it: the *update loop* (a later scan re-enters the backbone, is matched

against existing node identities, and applies a node-level upsert rather than a rebuild) and the *owner loop* (purple), in which the room occupant refutes, restores, and corrects labels, geometry, and attributes as highest-trust—but time-variant—evidence, each intervention landing as a provenance-carrying revision.

All main-campaign experiments observe one industrial room across three scan epochs of its working life: an exhibition-era *showroom* (T1), a *robot hall* test configuration (T2), and a controlled-change re-scan (T3); the same room appears as the “test room” in figure captions. The management console is a browser UI over the semantic store: it issues mission commands, shows the live graph and robot state, and gives the owner one-click refute/re-type/attribute edits whose writes every other consumer picks up immediately (Sec. 6.1).

The color separation in the architecture is a claim boundary, not a drawing convention. Everything in the backbone is seeded and parameter-fixed; re-running it on the same host reproduces every retained cluster byte-identically (measured in Sec. 5.2; across hosts we define reproducibility numerically, within floating-point tolerance). Everything downstream of the first VLM call is stochastic; there we claim *control*—schema-forced outputs, multi-view voting, automatic escalation, and confidence gating—rather than reproducibility.

Each object node carries the three TOSM description layers: a *symbolic* model (type, name, status), an *explicit* model (pose, extents, yaw, color), and an *implicit* model (movability, openability, key-object status). Verified and unverified content flow through the same graph but are separated at every interface: verification status is part of the node schema, gating is applied structurally at query time (Sec. 4.8), and the place layer is derived from verified objects only (Sec. 4.9).

The framework is fully automated *after registered TLS input*: scanner operation and vendor multi-setup registration are manual; everything from E57 to queryable graph—including escalation decisions and update classification—runs without human intervention, and objects the verifier cannot resolve remain explicitly unverified rather than being resolved by hand. Fig. 1 visualizes the resulting semantic map in the lineage’s explicit-model style [14], but with the segmented TLS points themselves rather than schematic boxes: place-layer regions form the floor, verified objects are colored by class, and unverified clusters render in gray—the confidence gate is visible in the map itself.

4. Methodology

This section describes the six processing stages of the framework in the order data flows through them: TLS preprocessing (4.1), geometric object decomposition with structure-remnant filtering (4.2), explicit modeling and provisional classification (4.3), multi-view VLM verification with automatic escalation (4.4), object identity matching (4.5), and confidence-gated graph upsert (4.6), followed by the structurally gated query interface (4.8) and the object-grounded place layer (4.9). Two further mechanism families—the corrective decomposition layers with the structure-condition ensemble, and the owner-evidence revision path—are defined in Sec. 4.7, with the failures that motivated them reported in Secs. 5.2 and 5.4; for brevity we refer to the geometric stages through explicit modeling as *Stage A* and to VLM verification as *Stage B*. Stages 4.1–4.3 form the *deterministic geometric backbone*: every random draw is seeded, so a re-run on the same host reproduces every retained cluster byte-identically (Sec. 5.2). Stages 4.4 onward are *probabilistic semantic inference*, controlled by voting, escalation, and confidence gating rather than by determinism claims. The pipeline is fully automated *after registered TLS input*: the only manual steps are operating the scanner and the vendor registration of multi-setup scans.

4.1. TLS Preprocessing

The input is a registered E57 point cloud from a survey-grade terrestrial scanner (Leica BLK360). When the export contains multiple setups, the loader applies each setup’s registered pose and merges them into a single cloud; a 70M-point six-setup scan and a

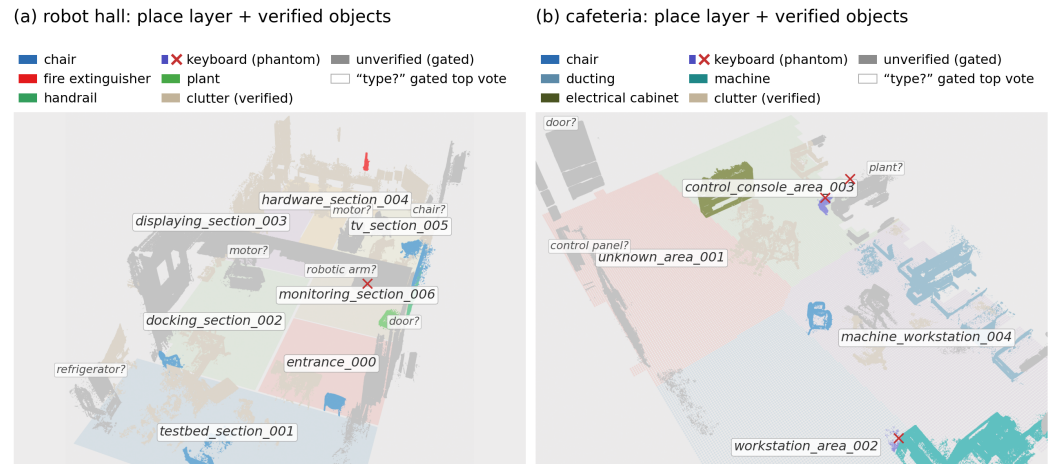


Figure 1. The generated semantic map, visualized at the verification stage (before owner feedback), so the confidence gate and the errors that pass it are both visible. (a) Robot hall: SLIC place-layer regions (pastel floor, named by the LLM ring-code scheme of Sec. 4.9) with the room-scoped object clusters rendered as their full-resolution segmented points, colored by verified class; gray clusters are unverified and demoted at query time. Gray italic callouts name the gate's withheld top votes—the hall's mobile manipulators surface as *motor?*, and both cafeteria doors are correctly suspected but held below the resolve threshold—while the hall's machines and wall TV sit unnamed inside the gray and clutter mass, the absorption class of Sec. 5.3. (b) The cafeteria stress scene under the same representation, its place layer derived by the identical SLIC recipe after synthesizing the occupancy grid from the scan's floor points (no SLAM run exists for this scene). Red crosses mark verified-keyboard clusters that are owner-refuted phantoms—no keyboard exists in either scene—and the olive “electrical cabinet” in (b) is in fact a planter partition, an owner-graded wrong label shown as the verifier produced it. Place naming is verifier-bounded accordingly: the cafeteria phantoms pull two region names toward workstation/console semantics, the same label-propagation mechanism as the *staircase_landing* case of Sec. 5.3.

pre-merged single-scan export are treated identically. The cloud is voxel-downsampled at 3 cm—dense enough to retain object detail while bounding the cost of all later stages. The cloud is kept in the scanner's registered frame; no global normalization is applied. Instead, the floor height is estimated where a stage needs it, by a robust statistic suited to that stage's input (the densest 5 cm height bin of the lower half-cloud for the remnant filter; the lower quartile of per-cluster bottom heights for the map-based filter of Sec. 4.7).

Structural surfaces are then removed by iterative RANSAC plane extraction [27]. Each round draws 300 seeded random point triples, keeps the candidate with the most inliers at a 5 cm distance threshold, and refits it once by a least-squares (SVD) fit to those inliers. A plane is removed only if it is structural by orientation: near-horizontal ($|n_z| \geq 0.85$; floors, ceilings, split ceiling levels) or near-vertical ($|n_z| \leq 0.2$; walls). The loop stops at the first of four conditions: (i) ten planes have been removed, (ii) fewer than 2,000 points remain, (iii) the best remaining plane supports less than 4% of the points still present—no large plane is left—or (iv) the best remaining plane is slanted ($0.2 < |n_z| < 0.85$), i.e., object-like, so removing it would begin carving objects. The fixed seed makes the extraction deterministic; the pre-removal cloud is retained as a *ceiling reference* for the remnant filter of Sec. 4.2 and for incremental re-runs. Fig. 3a,b show a scan before and after structural-plane removal.

4.2. Geometric Object Decomposition and Structure-Remnant Filtering

The structure-free cloud is partitioned into object candidates by fixed-parameter DBSCAN [29] ($\epsilon = 0.30$ m, 100-point minimum), followed by a *conservative giant-split*: any cluster whose horizontal footprint exceeds 2.5 m is re-clustered at a tighter $\epsilon = 0.20$ m, and

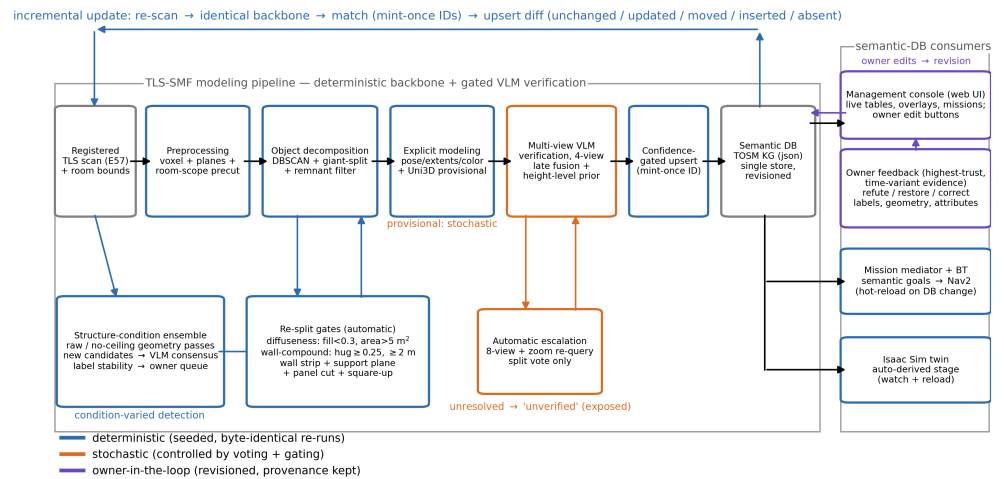


Figure 2. Framework architecture. Blue stages form the deterministic geometric backbone (including the automatic re-split gates and the structure-condition ensemble); orange stages are probabilistic semantic inference under voting, escalation, and gating control; purple is the owner-in-the-loop revision path. The knowledge graph is the single semantic store from which the console, the mission mediator, and the simulation twin are derived. Four loops: escalation (within verification), re-splitting (within decomposition), incremental update (across scans), and owner feedback (across revisions).

the split is accepted only if it yields at least two sub-clusters of 100+ points—separating merged neighbors without shattering genuinely large equipment. We position this stage deliberately: it is not a novel segmentation algorithm but a *reproducible decomposition designed for verification and incremental diffing*. Its value is that identical inputs yield identical clusters with stable ordering, which later stages exploit for result carry-over and change detection (Sec. 4.5). Fig. 3c shows the resulting instances, and Fig. 3d two of the TOSM records they eventually populate. The parameter sensitivity of this choice is quantified in Sec. 5.2. Four automatic corrective layers and one human-in-the-loop channel extend this conservative backbone; they are defined, with their trigger conditions and parameters, in Sec. 4.7.

Structure-remnant filter.

Real buildings violate the implicit assumption that structure is planar and objects are not. Our showroom scene has a three-level ceiling ($z = 2.48/2.18/1.95$ m): RANSAC removes each horizontal ceiling plane, but the *vertical soffit bands* at the level transitions survive removal and surface as phantom objects (Fig. 3e)—thin, wall-colored slabs that a downstream verifier confidently misreads (e.g., as wall-mounted keyboards or handrails, Sec. 5.3). We therefore filter any cluster that passes all of the following tests, with the thresholds fixed once and used unchanged on every scene. (i) *Soffit-scale geometry*: the cluster's vertical extent is at most 0.6 m and its bottom is at least 1.5 m above the floor. (ii) *Thin vertical planar band*: the best-fit plane (SVD) has $|n_z| \leq 0.25$ and an RMS off-plane thickness of at most 3 cm. (iii) *Ceiling above*: the cluster's footprint, padded by 0.3 m, contains at least 40 pre-removal points in the band from 2 cm to 45 cm above the cluster top, and (iv) *dense horizontal layer*: at least 60% of those points concentrate in a single ~ 7 cm slab around the modal 3.5 cm height bin—i.e., the band hangs from a real ceiling. (v) *Topmost layer*: the region above that slab (0.1–1.5 m higher) holds at most 20% as many points as the slab itself; window sills and wall remnants fail this test because wall continues above them. The filter reduced the showroom scene from 45 to 38 clusters and the robot hall from 88 to 81 while leaving every retained cluster byte-identical, so previously computed verification results remain valid. Horizontal and sloped ceiling patches are out of the filter's scope and are discussed as a residual limitation in Sec. 7.

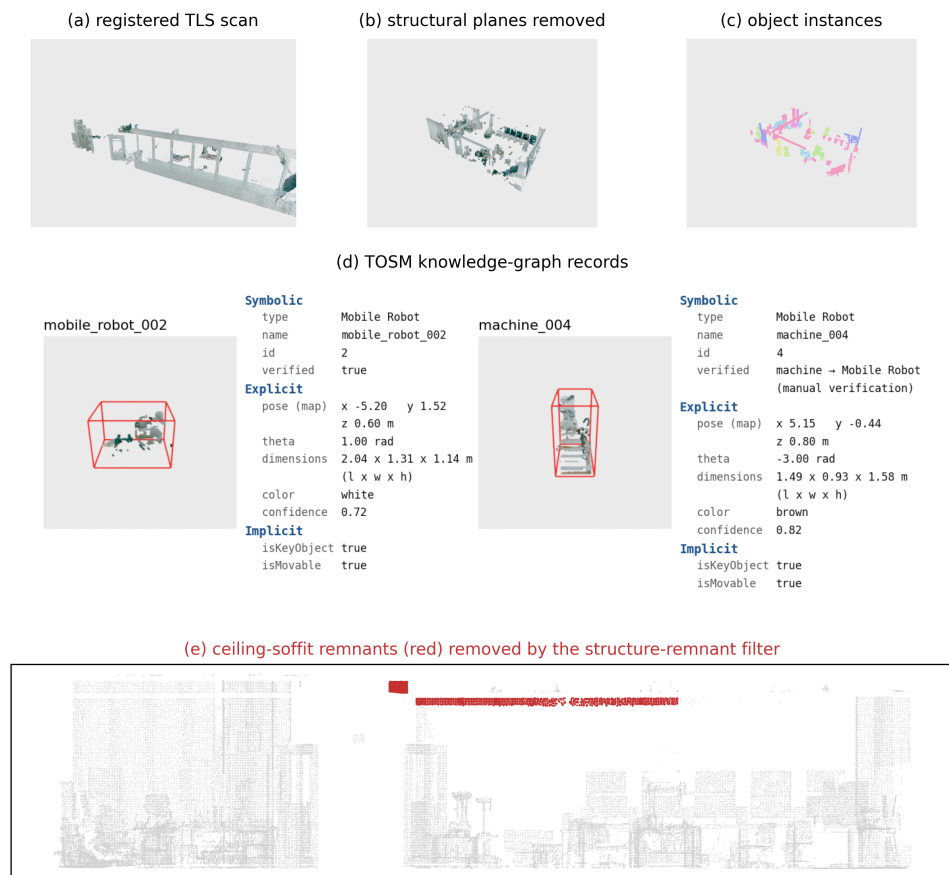


Figure 3. Geometric backbone of TLS-SMF, illustrated on the test-room scene. (a) Registered multi-station TLS scan after preprocessing. (b) The same cloud after seeded-RANSAC removal of structural planes (floor, walls, ceiling). (c) Deterministic decomposition of the residual cloud into object instances; colors denote cluster identity, and identical inputs reproduce identical clusters with stable ordering. (d) Two of the resulting TOSM knowledge-graph records with their symbolic, explicit, and implicit attributes; *machine_004* carries an owner-verified type correction (*machine* → *Mobile Robot*). (e) Structure-remnant filter, shown as a front elevation of the showroom scene: the vertical soffit bands at the three-level ceiling transitions (red) survive horizontal-plane removal and would otherwise surface as phantom objects; the filter removes them while leaving every retained cluster byte-identical.

4.3. Explicit Modeling and Provisional Classification

Each cluster is summarized by an explicit model: axis extents and centroid, yaw from the principal component of the horizontal point distribution, dominant color, and point count. These attributes are deterministic functions of the cluster; the backbone’s determinism claim covers geometry and these explicit attributes only—the provisional label below is stochastic and excluded from it. A provisional semantic label is then attached by Uni3D open-vocabulary classification over a 30-class industrial vocabulary. The provisional label is explicitly *not trusted*: our own vocabulary ablation shows that narrowing the vocabulary to the deployment domain *reduces* accuracy (Sec. 5.8). The provisional label serves only as VLM context and as a fallback name; every label a robot may act on must pass the verification stage below.

4.4. Multi-View VLM Verification with Automatic Escalation

Verification renders each object from four azimuths (0° , 90° , 180° , 270°) and submits each view independently to a vision–language model under a forced tool-use schema: the model must return a type, a confidence, and a stated reason, with the object’s measured

dimensions and provisional label provided as context. Free-text answers are impossible by construction, and prompts, decoding parameters, and model version are recorded with every result. The verifier model itself is a swappable component, not part of the design: the concrete model is an experimental control (Sec. 5.1), and Sec. 5.8 repeats the identical protocol across five commercial VLMs to characterize how the choice affects decisiveness, accuracy, and cost.

Per-view votes are combined by confidence-weighted late fusion. Let view v return label ℓ_v (lowercased; no synonym mapping is applied at fusion time) with confidence $c_v \in [0, 1]$. Each candidate label accumulates the *sum* of the confidences of the views that voted for it, $s(\ell) = \sum_{v: \ell_v = \ell} c_v$; the fused type is $\hat{\ell} = \arg \max_{\ell} s(\ell)$, and the *vote share* is its normalized confidence mass, $\sigma = s(\hat{\ell}) / \sum_{\ell} s(\ell)$. A confidence-weighted sum, rather than a view count, lets one hesitant dissenter matter less than a confident one. Three outcomes follow (Fig. 4). If the votes are unanimous, the object is *verified*; unanimity is empirically a strong signal (94.4–100% precision on the two scenes’ confirmed ground truth, Sec. 5.3), although Sec. 5.3 also documents its failure modes. If the vote splits, the object *escalates automatically*: eight additional images—four diagonal azimuths and four zoomed close-ups—are classified, and the fusion is recomputed over all twelve votes pooled (the base four included). If the post-escalation share reaches $\sigma \geq 0.6$ the object becomes *verified_escalated*; otherwise it remains *unverified*. The 0.6 threshold was fixed a priori, before any evaluation, as the smallest round value that demands a clear supermajority of the confidence mass: at 0.5 a twelve-view vote split between two labels of equal confidence could verify on a tie-break, whereas 0.6 requires the winning label to dominate all alternatives combined by a margin. The threshold gates *status only*—the fused label is recorded either way—so its effect is a precision–recall trade on the verified tier; Sec. 5.3 re-scores the stored votes across a threshold sweep to quantify that trade. Crucially, unresolved objects are not hidden, discarded, or resolved by a human: they stay in the graph with their failed candidate label attached, and the query interface (Sec. 4.8) is responsible for making their status impossible to ignore. Implicit robot-facing attributes (movability, openability, key-object status) are inferred once at fusion time from the winning view set.

Escalation bounds cost: only split votes pay for the wider view set (trigger rates 51–58%, Sec. 5.3), and per-call cost is $\sim \$0.009$, so a scene verifies for a few dollars (Sec. 7).

4.5. Object Identity Matching

The update problem is an *identity* problem before it is a geometry problem: a robot that plans against node IDs needs yesterday’s chair to be today’s chair. Our policy is **mint-once**: a node ID is minted when an object first enters the graph—derived from a fingerprint of its birth-time anchor (map, position, extent)—and never changes afterwards. The fingerprint is a candidate-retrieval key, not the identity itself; after minting, matching alone carries identity forward.

Matching between a new scan’s records and existing nodes runs in two passes:

1. **Spatial pass.** A record matches a node if their 3-D centroids agree within 0.5 m and each of the three extents (length, width, height) agrees within a per-axis tolerance of $\max(0.3 \cdot \max(d_{\text{old}}, d_{\text{new}}), 0.15 \text{ m})$ —a 30% relative band with a 15 cm absolute floor so that small objects are not over-constrained. No type constraint applies in this pass; a changed label on a geometrically stable object is recorded as an update, not a new object. Matches update pose and extents in place (unchanged / updated).
2. **Move pass.** Remaining records match remaining nodes of the *same type* with dimensions consistent under the same per-axis tolerance, within a 6 m horizontal search radius, producing moved nodes with preserved IDs. Generic types (*clutter*, *unknown*) are excluded from this pass: a generic label carries too little identity for displacement claims, and our real two-epoch data showed that admitting them fabricates 2–5 m “moves” between unrelated piles (Sec. 5.6).



Figure 4. Multi-view verification and automatic escalation, with per-view votes, fused share, and the three terminal statuses.

Conflict resolution. When several nodes satisfy the predicate for one record, the record claims the node with the smallest horizontal centroid distance; when several records compete for one node, the first claim wins and the node is removed from the candidate pool, so later records fall through to the move pass and then to insertion. Assignment is therefore greedy in record order rather than a global optimum—a deliberate simplicity choice: records arrive in the backbone’s deterministic cluster order, so the assignment itself is reproducible, and at the object densities of our scenes (≤ 0.6 objects/m²) multi-candidate conflicts are rare (Sec. 5.6 reports the observed conflict counts). The thresholds encode physical priors rather than tuned values: 0.5 m is below the smallest inter-object spacing we expect a scan-to-scan pose drift to produce for a stationary object (and is far above the ≤ 5 cm registration residual, Sec. 5.1); the 30%/0.15 m extent band absorbs partial-view truncation between scans without admitting differently-sized objects; and 6 m bounds a within-room relocation. They were fixed on the development epochs (T1/T2) and used unchanged afterwards.

Records that match nothing are minted as new nodes (inserted); nodes that no record claims are marked absent but retained with their full history, so an object that returns later resumes its original identity. Sec. 5.6 evaluates matching accuracy, identity-switch rate, and false-new/false-absent rates, including the occlusion level at which matching breaks (40% crop).

4.6. Confidence-Gated Graph Upsert

Each scan epoch applies its matched record set to the knowledge graph as a node-level upsert—no rebuild. Every node carries its verification status, provenance (per-view votes, fused share, escalation flag, model version), a per-revision history of every change, and firstSeen/lastSeen timestamps. Edges (spatial relations—*isNextTo*, *isOn*, *isAboveOf*—

place-scoped TOSM relation map (T3, rev 4): isInsideOf containment, intra-place object edges, place adjacency, and per-place key objects

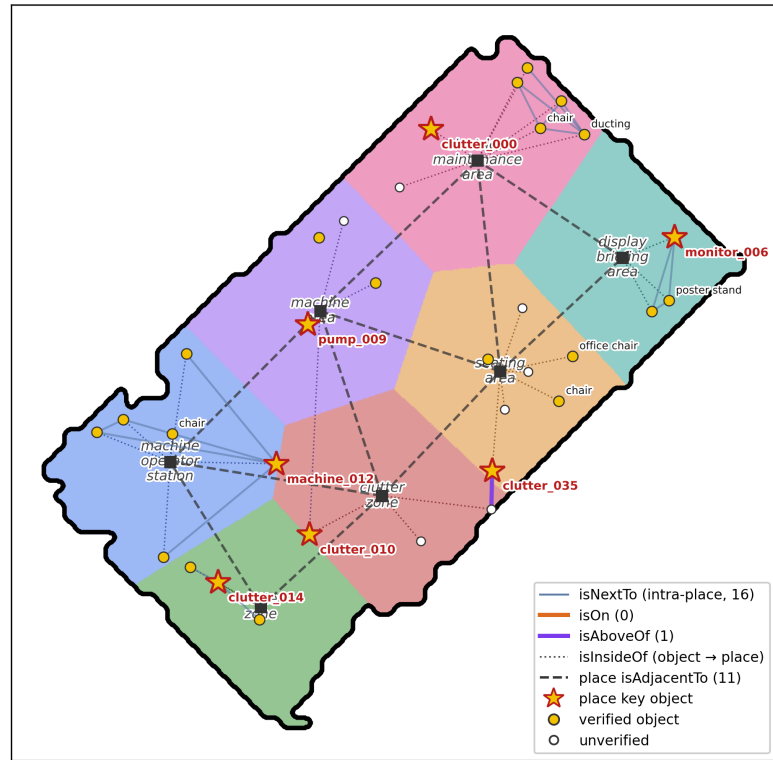


Figure 5. Place-scoped TOSM relation map (T3) over the place layer of Fig. 8 (pastel): dotted object→place isInsideOf containment, intra-place isNextTo edges, vertical isOn/isAboveOf relations, dashed place isAdjacentTo edges between region centroids, and per-place key objects (named stars). The map reflects the graph’s current revision, including owner feedback (Sec. 5.4).

recomputed from explicit models) are scoped to the map and revision, and organized by the place layer: isNextTo is kept only between objects of the same place (proximity across a place boundary is expressed at the place level as isAdjacentTo), vertical relations must pass a rotated-footprint overlap test (a separating-axis check that discards the long-range stacked relations oversized merged clusters otherwise produce), and each place designates a key object (the VLM’s implicit isKeyObject among verified members, falling back to the most confident specific-type verified member; objects refuted by owner ground-truth feedback are excluded from designation while remaining in the graph as annotated errors). On the T3 epoch this scoping removes 24 of 49 raw proximity edges—long cross-place spokes centered on the largest cluster—leaving 25 intra-place edges over 11 place adjacencies; Fig. 5 shows the resulting map. The result is exported both as a canonical JSON graph and as mapId-scoped Neo4j Cypher, so multiple sites and epochs coexist in one database. Because unchanged objects reproduce byte-identical clusters (Sec. 4.2), their verification results are carried over by content hash and an identical-input repeat scan costs zero VLM calls; a changed scan of the same acquisition quality pays only for changed clusters (3.1% of full cost in our controlled combined-edit measurement, Sec. 5.6). The carry-over is conditional on reproducible acquisition: when scan quality itself changes between epochs, as in our real T2→T3 transition, no cluster hashes match and the epoch pays full verification cost (Sec. 5.6). Fig. 6 summarizes the mint-once identity flow and the node outcomes of a re-scan.

Formal summary.

Consolidating the above: an object record is the tuple $r = (a, p, d, \ell, c, \tau, \pi)$ with birth anchor a (map ID plus position and extent quantized at 0.25 m, hashed once at first

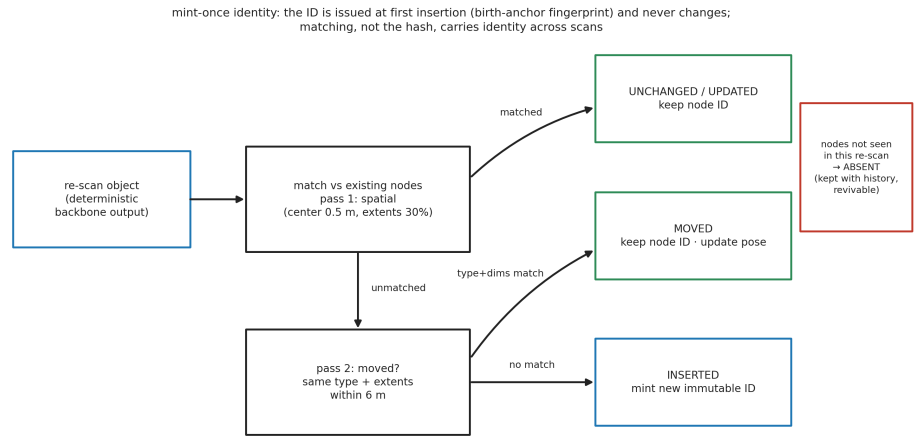


Figure 6. Identity matching and confidence-gated upsert: mint-once IDs, two-pass matching, and the five node outcomes across a re-scan.

insertion), pose $p \in SE(2) \times \mathbb{R}$, extents $d = (l, w, h)$, fused label ℓ , confidence c , verification tier τ , and provenance π (per-view votes, share, escalation flag, model version). The tier set is

$$\mathcal{T} = \{\text{verified, verified_escalated, verified_majority, verified_recovery, verified_owner, unverified, refuted}\},$$

ordered by evidence source: unanimity, escalated supermajority, agreement-gated recovery, and owner declaration (confidence 1.0); the query gate (Sec. 4.8) exposes only the verified tiers as typed objects. The match function is $m(n, r) = \mathbb{I}[\|p_n - p_r\|_{3D} \leq 0.5] \cdot \prod_{k \in \{l, w, h\}} \mathbb{I}[|d_{n,k} - d_{r,k}| \leq \max(0.3 \max(d_{n,k}, d_{r,k}), 0.15)]$, with the move-pass variant substituting a 6 m horizontal bound and an exact-type, non-generic constraint. The upsert decision for each record–node pair follows the deterministic cascade *match* → *move* → *insert*, with unclaimed nodes marked *absent*; every transition appends to the node’s history, and no node is ever deleted. All numeric constants above and in Secs. 4.1–4.6 were fixed on the development epochs (T1/T2) and applied unchanged to T3 and the cross-floor scene.

4.7. Corrective Decomposition Layers and Owner Feedback

The backbone of Sec. 4.2 is deliberately conservative, and its characteristic failures—diffuse residual webs, wall-flush compounds amputated by structure removal, structure noise that survives plane removal, and real objects absorbed into clutter—are corrected by five layers defined here. The first four are automatic and run inside the frozen pipeline on every epoch; the fifth is the human-in-the-loop channel. Each was *developed* in response to failures observed on the development epochs (Sec. 5.4 preserves that discovery narrative and its evaluation); their definitions, triggers, and parameters were frozen before the cafeteria evaluation and require no per-scene parameter adjustment at execution; held-out transfer is evaluated only where explicitly reported in Sec. 5.7.

1. **Diffuseness-gated re-split.** Trigger: a cluster whose 2-D occupancy fill (10 cm grid) is below 0.30 while its footprint exceeds 5 m²—real objects in our scenes fill 0.5–0.85. The cluster is re-decomposed by occupancy-core erosion: grid cells with ≥ 8 points and ≥ 3 occupied 8-neighbors seed connected components; components with ≥ 250 points survive. Compounds are further split at the support plane (the z-histogram peak 0.5–0.95 m above floor), with grounded cells spanning > 1.3 m vertically treated as legs/poles, and under-slab fragments re-attached at $\geq 60\%$ footprint overlap. An evidence gate (≥ 250 points, ≥ 5 m² or 12 m² for desk banks, ≥ 0.15 m height) keeps only physically plausible sub-objects.

2. **Wall-compound gate.** Trigger: $\geq 25\%$ of a cluster's points within 0.12 m of the exploration-grid wall line over a span of ≥ 2 m, with ≥ 250 points protruding off-wall, and the cluster not diffuse. The wall band is stripped, the support plane is detected as above, and the panel band is cut at density valleys along the wall, yielding unit-level decomposition of wall-mounted equipment. A final *panel square-up* pass then re-assigns every compound point that falls inside a panel unit's bounding box, padded by 6 cm horizontally and 5 cm vertically, back to that unit—drawing from the extracted support-post pieces and from points left unassigned by the cuts—so that post extraction and layer clustering do not leave bite marks or holes in the panel rectangles; it is instance-segmentation finishing with no semantic input.
3. **Structure-condition ensemble.** Trigger: every epoch. The geometric pass is repeated under three structure-removal conditions (none / ceiling-only / full); clusters found under an alternative condition but absent from the production set become new candidates, and only those are sent to verification. Cross-condition label agreement is stored per object as a `labelStability` attribute (stable/unstable) that prioritizes owner review; cross-condition *voting* is deliberately not used to fuse labels, because verifier biases proved correlated across conditions (Sec. 5.2).
4. **Hybrid grid-map structure filter.** Trigger: every epoch, after clustering. Each cluster is registered against the exploration occupancy grid (an independent, drift-free wall prior) and flagged by the first matching rule of a fixed cascade: outside-room ($>50\%$ of points beyond the filled room polygon), ceiling fixture (bottom >2.0 m above floor), wall remnant (wall-hug fraction—the fraction of cluster points within 0.15 m of the nearest grid wall line— >0.6 with PCA minor extent <0.22 m and a horizontal minor axis), or full-height wall band (z-span >2.5 m, per-metre-segment median thickness <0.3 m, along-wall length >2 m—the length condition spares wall-mounted panels). Flagged clusters are removed upstream of verification.
5. **Owner feedback and gated recovery.** The console exposes four operations, each applied as a normal KG revision with history: *declare* (sets `verified_owner` at confidence 1.0), *refute* (sets `refuted`, presence absent, and exclusion from key-object designation), *correct* (label or attribute in place, the VLM verdict retained in provenance), and *re-verify*. Recovery re-verification re-asks the verifier with `clutter` treated as *unidentified*, in two independent passes (open vocabulary; site vocabulary plus Uni3D candidates); a recovered label is adopted—as `verified_recovery`—only when the two passes agree under canonical synonym matching, or a pass agrees with the cluster's own base label, and the label is not `clutter`; a focused tie-break pass re-asks with exactly the two disputed labels and is accepted only when its answer is one of them at confidence ≥ 0.6 .

4.8. Query Interface with Structural Gating

The graph is queried through an LLM that receives a serialized view of the relevant nodes. The naive design—instructing the model not to assert unverified labels—fails in practice: in our benchmark the instructed model still asserted trap labels in 3 of 4 cases (Sec. 5.8). We therefore *gate by construction, not exhortation*: in the query-time view, every unverified node's type is demoted to unknown, and the failed candidate label is moved to a separate `failedCandidateType` field whose semantics (“a classifier guess that did not survive verification”) are stated in the schema, not in behavioral instructions. Fig. 7 contrasts the two record states as the query interface sees them. With the gate in the data rather than the prompt, the model cannot assert an unverified label without visibly contradicting its input: the four unverified-label traps are blocked in the gating-style ablation (4/4, Sec. 5.8), while in the end-to-end benchmark one count-phrased trap still leaks through the quarantine field itself (3/4, Sec. 5.5). Traps whose label was wrongly *verified* lie beyond any status-based gate—a bound we quantify in Sec. 5.5.

In interaction, the gate makes confidence part of the answer rather than metadata. Asked “is there a keyboard in the room?”, the system answers from the gated view: a

(a) verified object record	Symbolic <pre> type control panel name control_panel_007 verified true (VLM 4-view unanimous) voteShare 1.0 </pre> Explicit <pre> pose x -2.24 y -3.54 z 1.00 m (theta 3.03 rad dims 1.87 x 0.43 x 2.83 m color brown conf. 1.00 </pre> Implicit <pre> keyObject true movable false </pre>	(b) unverified record (gated)	Symbolic <pre> type unknown (gated; was name 'keyboard') name keyboard_046 verified false - demoted at query time voteShare 0.281 </pre> Explicit <pre> pose x 8.59 y 4.22 z -0.58 m theta (map) theta 1.67 rad dims 0.58 x 0.06 x 0.51 m color brown conf. 0.28 </pre> Implicit <pre> keyObject false movable true </pre>
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Figure 7. TOSM semantic-object records (symbolic / explicit / implicit layers). (a) A verified record asserts its type. (b) An unverified record is structurally gated at query time: the type is demoted to **unknown** and the failed classifier guess is quarantined in **failedCandidateType**.

verified keyboard is asserted with its location, while an unverified one yields “an object at (x, y) was proposed as a keyboard by the classifier but did not pass verification”—an answer a robot can act on conservatively (route a person, lower approach speed) instead of a confident guess.

4.9. Object-Grounded Place Layer

Robot tasking refers to places (“the docking area”) as often as to objects. Following the TOSM place schema of DK-SMF [15]—boundary polygon, containment, and purpose—we add a place layer grounded in the verified object layer. Place regions are obtained automatically from the robot-facing occupancy grid map (0.05 m—generated from the scan here, and equally the exploration SLAM map on deployment), following DK-SMF’s recipe: the room interior delimited by the mapped outer boundary is partitioned by SLIC superpixel segmentation [42] with the number of superpixels as the governing parameter (7 requested here), fragments below an area threshold (1.5 m²) are discarded, and each region’s centroid is the moment-based center of its cells (cf. grid-map room segmentation [43, 44]). Each object’s `isInsideOf` resolves to the region containing its anchor cell. Each place is then summarized by its *ring code*: the sequence of member objects sorted by bearing around the place centroid—restricted to verified-tier objects, so the confidence gate propagates into place semantics—and a single LLM call names the place from its ring code. Because the segmentation depends only on the map, region identity is stable across scan epochs, and the same regions re-derive names that track the room’s actual use (the machine bay: exhibition-era `machine_access_area` → storage-era `clutter_zone` → test-era `machine_operator_station`; Sec. 5.6), demonstrating that place semantics stay current without manual re-annotation. Fig. 8 shows the resulting T3 place layer. The robot layer of TOSM is out of scope here.

5. Experiments

5.1. Setup

Scenes and epochs.

All main-campaign data comes from BLK360 scans of two industrial indoor environments that are, in fact, one room observed across its working life—a property we exploit deliberately. The *showroom* epoch (June 2026) captures the room in an exhibition configuration, scanned manually with the door open; its semantic target is the wall-removed export (38 objects after remnant filtering). The *robot hall* epoch (July 2026) captures the same room reorganized as a robot test hall, scanned by autonomous visibility-driven exploration (81 objects, room-scoped to 55). A third epoch (late July 2026) adds a controlled-change re-scan at 99.6% coverage (Sec. 5.6). Because the two primary epochs differ in layout, furnishing, and scan modality, they function as independent scenes for per-epoch evaluation (Secs. 5.2–

object-grounded place layer (T3): SLIC superpixel regions over the nav-stack grid map;
LLM-derived names from verified-object ring codes

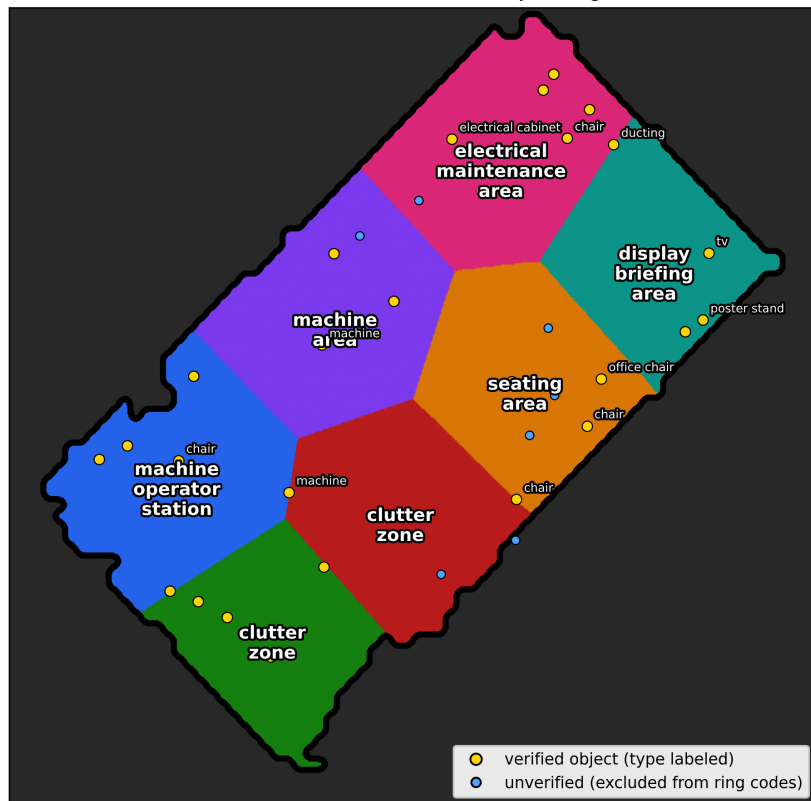


Figure 8. Object-grounded place layer at T3. SLIC superpixel regions [42] tile the room interior delimited by the nav-stack grid map’s outer boundary and carry LLM-derived names computed from verified-object ring codes; unverified objects (blue markers) do not contribute, so the confidence gate propagates into place semantics. Shown at the graph’s owner-corrected revision (Sec. 5.4).

Table 3. Scene statistics (deterministic detection “det” sets, after structure-remnant filtering). T1 semantic processing starts from a manually wall-removed export (4.9M points) of the 46.3M-point raw scan.

	Showroom (T1)	Robot hall (T2)	Re-scan (T3)
Scan mode	manual, 1 setup	exploration	exploration (SOTA), 6 setups
Raw points	46.3M	68.6M	70.2M
Clean cloud (post-removal)	—	267K	294K
Objects (filtered)	38	81	79
Room-scoped	38	55	41
Removed planes	10	10	10

5.5) while enabling a real multi-epoch update study (Sec. 5.6). Scene statistics are given in Table 3. A separate scene—a carpeted cafeteria with sofa booths and vending machines on a different floor of the same building, scanned in August 2026 (57M points, 28 clusters)—is held out of the main campaign and used only as a cross-floor, out-of-vocabulary stress case in Sec. 5.8. Both sites are photographed in Fig. 9.

Ground truth.

Ground truth was established in two passes: a render-based audit of every cluster, followed by confirmation from the room’s owner. Labels are graded *confirmed* (owner-verified), *plausible*, or *unknown*; accuracy is reported against confirmed GT (with lenient variants including plausible where stated), unknown objects are excluded, and structure

(a) test room, 4F (photographed in the robot-hall configuration)



(b) cafeteria stress scene, 8F



Figure 9. The two experimental sites. (a) The 4F test room, photographed in its robot-hall configuration (the showroom epoch T1 is an earlier furnishing of the same room; visible: the AMMR platforms, the wall-mounted monitor bank of Sec. 5.2, and the disinfection robot). (b) The 8F cafeteria stress scene of Sec. 5.8: fabric sofas, planter partitions, the vending row, and—center right—the brown door recovered by escalation on full-resolution renders.

Table 4. Evaluation-set lineage: how every cluster set used in Sec. 5 derives from its scan. All counts are recomputed from the archived stage outputs in the data package; † marks the one count grounded only in the experiment log. “conf.” = GT-confirmed.

Lineage	Stage flow (cluster counts)
Showroom 57-set (Sec. 5.3)	30 pre-split → 57 split → 50 after soffit filter → 50 GT-scored (36 conf.)
Showroom det set	45 raw → 38 after soffit filter → 38 room-scoped
Robot hall det	88 raw → 81 filtered → 55 room-scoped → 51 GT-scored (32 conf.)
T3 det1	79 merged → 41 room-scoped (= owner audit set) → 26 filter survivors
T3 det2 (precut)	66 → 35 room-scoped → 27 to VLM (8 wall sheets excluded)
T3 det3/det4	33 consolidated → 46 [†] adjudicated
Cross-epoch KG (present)	rev7→14: 47, 49, 51, 63, 64, 64, 46, 46 (current)
det5 frozen re-run (Sec. 5.7)	66 → 35 in-room, scored vs. the 46-object graph
Cafeteria 8F	28 → 25 GT-scored (20 conf.)

artifacts are tracked separately. Two showroom sets must be distinguished: Table 3 counts the 38-object reproducible det set, while the verification study (Sec. 5.3) uses the original 57-cluster decomposition of the same scan, of which 50 clusters are GT-scored after excluding 7 structure artifacts—yielding confirmed 36 / plausible 12 / unknown 2. Table 4 connects every evaluation set used in this section—the 38-object det set, the 50-scored showroom subset, and the T3 41-, 35-, and 46-object sets—to its derivation stage, so no denominator has to be inferred from prose. For the room-scoped robot hall (55 objects), GT covers 51 clusters: confirmed 32 (lenient $n=42$) after owner corrections. The cafeteria stress scene follows the same practice: 25 of 28 clusters scored (3 structure artifacts excluded), confirmed 20 / plausible 5. Both audit passes share an anchoring channel—the render audit presents proposed labels that the owner then corrects—so confirmed grades are owner decisions over physically known rooms rather than blind annotations; the owner overturned model-favored labels in several cases (a “refrigerator” re-graded as a banner, a verified keyboard

refuted as a phantom, “motors” re-identified as mobile manipulators), which bounds but does not eliminate this bias. To measure it directly, a second annotator—a co-author familiar with the facility who had seen neither the T3 pipeline output nor the owner audit—labeled all 41 T3 clusters *blind*, from full-resolution multi-view render sheets carrying only neutral shuffled IDs, in free vocabulary with a 1–5 confidence. On the 33 clusters both annotators labeled, the blind labels agree with the owner’s on 31 (93.9%; the two disagreements are low-confidence blind labels, 2–3/5); the blind annotator abstained on 5 clusters and both were uncertain on 2. The most informative split for anchoring is by whether the owner had *overridden* the model: on the 31 clusters where the owner overturned the pipeline label, the blind annotator independently reproduced the override on 25 of the 26 it scored (abstaining on 5), versus 6 of 7 where the owner had accepted the model label. The 25/26 reproduction of the owner’s overrides argues against those corrections being driven primarily by model anchoring; the accepted subset is small (6/7), so residual anchoring cannot be excluded, and we therefore describe the owner ground truth as anchoring-bounded rather than anchoring-free. The blind kit, answers, and scoring are in the data package.

Metrics.

Classification accuracy is top-1 against GT under a documented synonym-tolerant matching (e.g., *mobile manipulator* $\approx$ *robotic arm*). Verification quality is reported as per-tier precision (unanimous, escalated, unverified). For query evaluation, a *hallucination* is an unsupported atomic factual claim, and the rate is unsupported claims over all factual claims in the answer set. Update quality uses the metrics of Sec. 4.5: matching accuracy, identity-switch rate, false-new, false-absent, and unchanged stability.

Controls.

The verifier for the main campaign (Secs. 5.2–5.6) is fixed to a single model, *claude-sonnet-4-6* (the frontier mid-tier at campaign start): every verification, escalation, and owner interaction in the accumulated knowledge-graph lineage was produced under this one verifier, so results remain comparable across sections and revisions. The verifier choice itself is then evaluated in Sec. 5.8, which repeats the identical Stage-B protocol across five commercial VLMs and re-runs the headline verification study under the best-ranked one. All query conditions share the same LLM (*claude-sonnet-4-6*), system prompt, query set, decoding parameters, retry policy, and forced answer schema; conditions differ *only* in the knowledge view they receive. The LLM-only condition—an *unverified Stage-A LLM* baseline—receives the same scene’s Stage-A object dump, so it is not information-starved—it is verification-starved. Threshold provenance is stated in two parts. Constants in the base pipeline (Secs. 4.1–4.6) were fixed on T1/T2 and applied unchanged to T3 and the cafeteria. The corrective-layer thresholds of Sec. 4.7 were developed on the test-room lineage—including failures revealed by the T3 owner audit (Sec. 5.4)—and were then frozen before evaluation on the cafeteria stress scene; accordingly, T3 is treated as a retrospective development-scene replay of those layers rather than a held-out evaluation (Sec. 5.7). Nothing is trained, so the split that matters is the development room (T1/T2/T3) versus the held-out cafeteria, and Sec. 5.7 states which claims survive on which side of it.

Generative AI use.

Generative models enter this study in two strictly separated roles, disclosed here per the journal’s GenAI policy. As *experimental components under test*: commercial VLM/LLM services perform Stage-B multi-view verification (*claude-sonnet-4-6* throughout the main campaign; Anthropic PBC, San Francisco, CA, USA), the query-benchmark answering under all four conditions (the same model), place-name generation from ring codes, and the five-verifier comparison of Sec. 5.8 (Anthropic and OpenAI models); every such call is made under a forced tool-use schema with prompts, decoding parameters, model identifiers, and per-call usage archived in the data package, and these models contributed no text to this article. As a *writing aid*: a generative AI assistant was used for language editing and LaTeX

formatting of the manuscript, as declared in the Acknowledgments; it played no role in study design, data collection, or analysis beyond the archived experimental calls above.

5.2. Segmentation: Repeatability, Filtering, and Sensitivity

Repeatability.

Running the full pipeline twice from the same E57—including ten-plane structure removal—reproduces every cluster byte-identically on both scenes (hash-verified). To probe robustness beyond one software stack, we packaged the backbone into a version-pinned container and re-ran the robot-hall scan under a deliberately different environment (glibc 2.41 vs. 2.39, Python 3.12.14 vs. 3.12.3): all 81 retained clusters remain byte-identical to the native run, and consecutive container runs reproduce each other exactly. Running the same container on a second machine with a different CPU vendor and microarchitecture (Intel Core i9-13900H laptop vs. the AMD Ryzen 9 9900X reference workstation) reproduces the identical per-cluster checksums and aggregate hash—the determinism claim therefore holds across operating environments and x86-64 CPU vendors under the pinned library set. The verification kit (container recipe, reference checksums, comparison script) ships with the data package so the check can be repeated elsewhere; bit-level reproducibility across instruction-set architectures is not claimed. This is the property the incremental machinery builds on: identical clusters mean carryable verification results and attributable diffs.

Remnant filter.

The structure-remnant filter of Sec. 4.2 removed all fourteen owner-confirmed ceiling-soffit phantoms (seven per scene: showroom 45→38, robot hall 88→81) while leaving every retained cluster unchanged. Before the filter, these phantoms did not merely inflate the object count: they produced confidently verified fictions (keyboard panels, handrails) that survived multi-view voting, since every view showed the same wall-colored slab. Detecting structure remnants *geometrically*, upstream of semantics, proved more reliable than asking the verifier to doubt itself (cf. the pose-context ablation, Sec. 5.8).

Parameter sensitivity.

A 36-combination sweep (giant-split footprint $\times$ DBSCAN $\epsilon \times$ minimum points) shows a stable plateau at footprint 2.0–2.5 m: instance recall degrades at 3.0 m as neighbors merge and at 1.5 m as equipment shatters, while ϵ /min-point extremes collapse recall to 0.21. Because scoring a sweep against the operating point’s own decomposition would be circular, we re-scored every combination against an independent object-level reference: the owner-audited inventory, in a membership-independent form (all clean-cloud points falling inside each owner-audited oriented box, so no clustering decision defines the reference). On the robot hall (42 reference objects) the conclusion survives intact: the min-points-100 band is a broad plateau, and the operating point (2.5 m / 0.30 / 100) sits at the grid maximum under the independent reference (recall 0.952, a uniform ~ 0.05 below the cluster-referenced numbers, with the plateau shape unchanged). On the showroom, the audit surfaced an error we report plainly: the original showroom sweep rows had joined the reference onto the operating run’s own clusters, and against the genuinely audited 57-object inventory *no* grid combination of the base clustering exceeds recall 0.33 — the audited granularity of that scene is produced by the split-refinement stage, not by any base-DBSCAN setting. The showroom rows are therefore evidence for the necessity of the refinement stage, and the parameter-robustness claim rests on the non-circular robot-hall table (full grids in the data package).

Conditional Point-SAM refinement.

Under a pre-specified rule (adopt iff $\geq 50\%$ of merged-cluster failures separate correctly and label accuracy improves ≥ 5 points), Point-SAM separated 3/6 flagged merged clusters and sub-cluster labels reached 9/11 versus 5/11 for their parents (+36.4 points, Fig. 10), so the rule is met for *flagged merged candidates*; because automatic candidate selection also

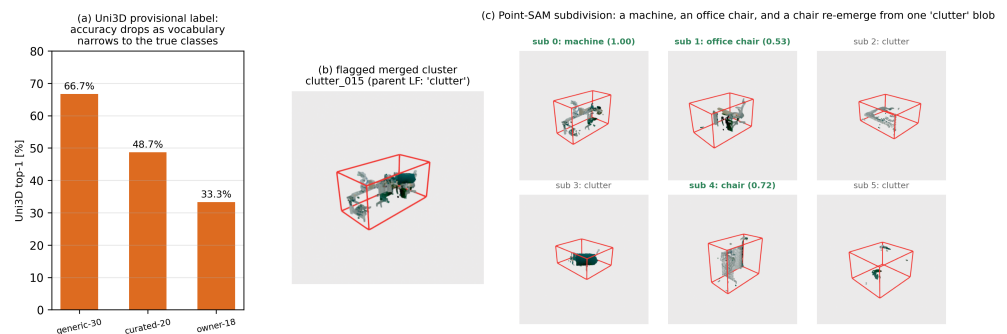


Figure 10. Learned components used as bounded tools. (a) The Uni3D provisional label degrades as the vocabulary narrows to the deployment’s true classes (Sec. 5.8)—why it is treated as untrusted context. (b,c) Conditional Point-SAM refinement of a flagged merged cluster: a machine, an office chair, and a chair re-emerge from a single `clutter` blob, each re-verified independently.

Table 5. Segmentation $\times$ classification backbone comparison on a wall-removed industrial TLS scene. The metrics are screening proxies (no point-level GT) and do not measure instance quality; the PTv3 row is a semantic-segmentation head whose outputs are not instance-comparable with the clustering rows and is included only to document why closed-set semantic heads were screened out. Instance quality of the retained backbone is validated independently in Secs. 5.2 (owner-referenced sweep) and 5.7 (end-to-end recall).

Pipeline	Coverage	Unnameable	Classes	Dominant failure
DBSCAN + Uni3D	98.3%	23.3%	13	unlabeled coverage
SPFormer + Uni3D	51.1%	45.9%	9	furniture bias (chair 67/88)
PTv3 (semantic)	—	—	17/20 emitted	46% “wall” after removal
DBSCAN + PointCLIP V2	98.3%	—	1–7	collapse (pipe 23/30)

produced 2/6 false positives that fragment single objects, Point-SAM is retained only as an optional, manually triggered diagnostic refinement—it is not part of the frozen pipeline of Sec. 4.7 and contributes nothing to the end-to-end replay of Sec. 5.7.

Coverage stress case.

The T3 epoch supplied a stress case: at 99.6% coverage, inter-object gaps close and DBSCAN chains 120K points (41% of the clean cloud) into one 10.4×9.6 m mega-cluster that survives the giant-split. Refinement recovers little of the resulting diff inflation—of the epoch’s 45 absent nodes, 43 retain a source cluster to re-associate against; Point-SAM ($K=7$ prompts) re-associates 4 of these 43, tighter re-splitting ($\epsilon=0.12$) 5 of 43—which is itself informative: $\sim 88\%$ of that epoch’s absences reflect genuine scene change, not segmentation artifacts (Sec. 5.6).

Backbone selection.

The DBSCAN + Uni3D backbone was chosen against learned alternatives on a wall-removed industrial scene (122.7K points), summarized in Table 5. The metrics are deliberate proxies (no point-level ground truth exists for this scene): point coverage, the fraction of objects the open-vocabulary classifier cannot name (a boundary-quality proxy), and label diversity. ScanNet-pretrained SPFormer hallucinates furniture on industrial content (67 of 88 instances “chair”) at half the coverage; PTv3’s closed-set semantic head labels 46% of a scene *whose walls were already removed* as wall — a closed set cannot abstain; and PointCLIP V2 collapses to a single class. The geometric backbone wins not by being smarter but by refusing to guess: its failure mode is unlabeled coverage, which the verification and owner layers are built to absorb. Point-SAM enters only as the conditional refiner above.

Table 6. Verification accuracy against confirmed GT, and per-tier precision. Accuracy rows are over confirmed GT ($n=36/32$); trigger rates and tier counts are over all GT-scored clusters (50 showroom / 51 robot hall), so, e.g., 21 unanimous + 29 split = 50.

	Showroom ($n=36$)		Robot hall ($n=32$)	
	base-4	escalated	base-4	escalated
Accuracy	77.8%	88.9%	78.1%	84.4%
95% Wilson CI	61.9–88.3	74.7–95.6	61.2–89.0	68.2–93.1
Recovered / regressed	5 / 1		3 / 0	
Escalation trigger	58% (29/50)		51% (26/51)	
Unanimous precision	100% (19/19; 21 unan.)		94.4% (17/18; 25 unan.)	
Escalated precision	100%		88.9%	
Unverified precision	60%		40%	

Diffuseness-gated re-splitting.

This layer and the next are defined, with triggers and parameters, in Sec. 4.7; here we report what they did. Two T3 clusters that survived every upstream filter are *diffuse webs*: 95K and 13K points spread over 80 and 12 m² at a 2D occupancy fill below 0.3, versus 0.5–0.85 for every real object in the scene — a clean geometric signature, but one density clustering cannot act on, because the webs are connected at ~ 1.8 cm point spacing at any usable ϵ . A two-stage re-split resolves them by *cause* instead of size: points hugging the exploration-grid wall line (inside the *room polygon*, so wall-flush furniture survives the test that free-space masks would fail) are structure noise ($\sim 70\%$ of the ~ 95 K-point residual); out-of-room bleed-through accounts for another 7%; the remainder is re-clustered by occupancy-core erosion and, for compounds, split at the support plane detected as the z -histogram peak 0.5–0.95 m above floor. The owner’s ground truth for the largest compound — “a robot arm on a desk, part of a mobile robot, wall noise” — is recovered exactly: the arm (consensus-verified), the desk (slab plus under-desk storage), and the partially scanned mobile robot, plus three columns and a plant. Applied to a 6.2 m wall-flush cluster that verification had confidently labeled *electrical cabinet* (0.83), the same machinery — wall-band strip, support plane, then density-valley cuts along the wall in the panel band — recovers the owner’s inventory at unit granularity: nine of the ten owner-confirmed monitor units, the desk beneath them, and, from the off-wall remainder, a wheelchair robot and a single table leg. Unit boundaries are approximate — the dark screens return few points, so the cuts follow density valleys in the surrounding wall band — and the result is claimed as unit-level recall, not precise per-unit segmentation. The gate turns “too diffuse to be an object” from a discard decision into a decomposition trigger.

Structure-condition ensemble.

Structure removal is itself a segmentation parameter: it amputates wall-flush objects along with the wall. Running the geometric pass under three removal conditions (none / ceiling only / full) and fusing the results recovered two multi-metre tables that exist in *no* single-condition segmentation of the production path — they appear only when walls are left in place — raising in-room recall by four real objects ($\sim 8\%$) as a standalone pass (its incremental gain in the cumulative stack of Table 10 is smaller because the re-split and wall-compound layers already recover part of the same set). The fusion also quantifies label fragility: only 10 of 37 re-found objects keep the same label across conditions, and cross-condition *voting* corrected nothing, because the verifier’s biases are correlated across conditions (both recovered tables were unanimously mis-verified as conveyor belts in every condition, while the room’s actual conveyor unit stayed *clutter* in every condition). The ensemble is therefore used as a recall tool plus a per-object `labelStability` attribute that prioritizes owner review — not as a label fuser.

5.3. Verification: Fusion and Escalation

Escalation lifts confirmed-GT accuracy from 77.8% to 88.9% on the showroom and from 78.1% to 84.4% on the robot hall (lenient: 76.2% $\rightarrow$ 83.3%, $n=42$), recovering 5 and 3 split-vote objects while regressing 1 and 0—a consistent observed improvement across both configurations, though at these sample sizes the paired contrast does not reach conventional significance (exact McNemar on recovered/regressed: $p=0.219$ and $p=0.25$), which the Wilson intervals of Table 6 make explicit (Table 6; Fig. 12 shows the accuracy–cost trade-off at $\sim \$0.009/\text{call}$; escalation adds 232 and 208 calls per scene). The tier structure justifies gating: unanimous and escalated tiers reach 88.9–100% precision, while the unverified residue is 40–60%—exactly the content the query interface must demote. These precision figures are scoped to confirmed GT: on the showroom, one unanimous vote falls on a *plausible*-graded cluster and is wrong there (a suspected mobile robot absorbed as *clutter*), so the tier’s precision over confirmed+plausible GT is 20/21 rather than 19/19—the scoping convention is reported so that the headline is not read as a general property of unanimity (the stress scene of Sec. 5.8 degrades it substantially). A repeat of the showroom run under identical settings bounds the verifier’s run-to-run variance: base-4 accuracy reproduces exactly (77.8%), escalated accuracy shifts by a single object (86.1% vs. 88.9%), the trigger rate by two points (60% vs. 58%), and unanimous and escalated precision hold at 100% in both runs—a ± 3 -point scale against which the verifier comparisons of Sec. 5.8 should be read. A separate check evaluates the room-scoped T1 objects in the update-study frame (Sec. 5.6), where showroom GT is transferred to the re-decomposed clusters by bidirectional point overlap; on the resulting small confirmed set ($n=8$), verified-tier precision is 100% with 7/7 unanimity.

Verification yield tracks scan quality: the manual scan leaves 47% of objects unverified versus 24% for exploration scans, and re-processing the T3 scan from a single setup (1/6 coverage) drops the verified tier from 75.6% to 56.6% on identical geometry. Coverage buys verifiability.

Per-tier recall.

Precision alone understates what gating costs, so Table 7 completes the tier picture with recall and F_1 against confirmed GT. The pattern is consistent: the combined verified tier reaches recall 0.722/0.781 on the two industrial scenes (F_1 0.839/0.847) but only 0.250 on the cafeteria, where the gate abstained on half of the confirmed objects—the stress scene’s segmentation failures surface here as recall, not as precision. The escalated tier is small everywhere (recall 0.194–0.250), consistent with its role as a recovery channel rather than a primary path.

Threshold sensitivity.

Because the 0.6 share threshold gates status only (Sec. 4.4), its effect can be quantified exactly by re-scoring the stored votes at alternative thresholds—no new verifier calls. Sweeping 0.40–0.90 in steps of 0.05 shows 0.6 sitting on a plateau rather than a cliff. On the showroom, thresholds 0.55 and 0.60 produce identical tiers (escalated-tier precision 100%); lowering to 0.50 trades combined verified-tier precision (1.000 $\rightarrow$ 0.966) for recall (0.722 $\rightarrow$ 0.778), and raising to 0.70 keeps precision saturated while dropping recall to 0.639. The robot hall behaves the same way (at 0.60: escalated precision 0.889, recall 0.781; at 0.70: precision 1.000, recall 0.688), and on both industrial scenes F_1 is maximized on a broad 0.45–0.65 band with every 0.05 step moving at most one or two objects. The stress scene shifts the whole curve downward without changing its shape—the threshold cannot repair render- or segmentation-level failures (Sec. 5.8)—which is the scene dependence noted for the tiers themselves. Between 0.50 and 0.70 only 11/34, 12/49, and 7/19 escalated objects change tier on the three scenes, and those bands mix correct and incorrect labels almost evenly, which is why no alternative threshold dominates 0.6 by more than single-object noise. We therefore retain the a-priori 0.6 rather than tuning it post hoc.

Table 7. Per-tier precision, recall, and F_1 on confirmed ground truth (synonym-tolerant). Recall counts a confirmed object as retrieved only if the tier contains it with the correct type; the unverified row is the abstention tier, so its “precision” is only the fraction of abstained objects whose withheld top vote happened to be correct.

Scene	Tier	n	n_{conf}	Prec.	Rec.	F_1
Showroom	verified (unanimous)	21	19	1.000	0.528	0.691
	verified_escalated	8	7	1.000	0.194	0.326
	combined verified	29	26	1.000	0.722	0.839
	unverified (abstained)	21	10	0.600	0.167	0.261
	confirmed GT total	50	36			
Robot hall	verified (unanimous)	25	18	0.944	0.531	0.680
	verified_escalated	16	9	0.889	0.250	0.390
	combined verified	41	27	0.926	0.781	0.847
	unverified (abstained)	10	5	0.400	0.062	0.108
	confirmed GT total	51	32			
Cafeteria	verified (unanimous)	8	7	0.571	0.200	0.296
	verified_escalated	6	3	0.333	0.050	0.087
	combined verified	14	10	0.500	0.250	0.333
	unverified (abstained)	11	10	0.200	0.100	0.133
	confirmed GT total	25	20			

Error taxonomy.

Four recurring error classes emerged (Fig. 13): (a) *structure-remnant phantoms*—soffit bands verified as keyboards; eliminated upstream by the remnant filter; (b) *unanimous-but-wrong*—a hanging chain unanimously verified as a door handle; an escalated keyboard the owner refuted; partition fragments as chairs; at T3, a residual 7.4 m wall segment that survived the remnant filter was escalation-verified as a staircase—owner-refuted (“just the wall”)—and its phantom label then propagated into a place name (staircase_landing, Sec. 4.9), showing that place naming is verifier-bounded too and that wall-scale remnants can slip past a filter tuned for ceiling-soffit bands (cf. Sec. 7). The same propagation reaches the relation layer: a keyboard verified at full confidence in a room that contains no keyboard (owner-refuted) had been auto-designated its place’s key object before the refutation demoted it—key-object designation therefore honors owner feedback (Sec. 4.6). A third refutation is the mirror image: the room’s large wall-mounted TV was geometrically detected (monitor_006, 2.5×1.8 m) yet left unverified at 0.26 confidence, while a small co-located cluster was escalation-verified as a fire extinguisher (0.72) the owner refuted—the salient real object failed the gate a phantom passed. Unanimity is strong evidence, not proof, and the gate inherits the verifier’s blind spots; (c) *clutter absorption*—a disassembled robot, a poster stand, and a low-reflectance floor fan (674 points) each unanimously verified as clutter: geometrically detected, semantically flattened; (d) *render ambiguity* → unverified, the intended failure mode. Classes (b) and (c) motivate reporting verification as *tiered evidence* rather than truth, and drive the structural gating design.

Class-level breakdown.

Table 8 disaggregates every scene’s ground truth into class families and per-family verification outcomes, making the error classes above quantitative. Three patterns stand out. First, the vocabulary gap is class-structured: robots are the worst family everywhere they occur (0/4, 0/3, and 0/8 correct across the three robot-bearing sets)—absorption, not misperception, since most land in clutter or abstain. Second, displays and fixtures abstain heavily on the showroom (0/5 and 0/6 correct, all withheld or mislabeled)—flat panels and ceiling fixtures are exactly the render-ambiguous shapes the taxonomy predicts. Third, the T3 owner audit shows the same distribution end-to-end (6 correct / 25 wrong / 10 abstained over 41), confirming that per-family results, not the aggregate, are the actionable signal for vocabulary and geometry work.

Table 8. Ground-truth object classes per scene, grouped into families, with each object’s verification outcome in the canonical run (synonym-tolerant matching as in the escalation analysis). “Filtered” = excluded before scoring (structure artifact or out of room scope). For the T3 owner audit, correctness follows the owner’s category labels; one verified cluster with no owner answer is counted as wrong in the total row only.

Scene	Family	GT	Corr.	Wrong	Unver.	Filt.
Showroom (57-set)	seating	5	4	1	0	0
	robots	4	0	1	3	0
	displays/monitors	5	0	0	5	0
	machines/electrical	6	3	0	3	0
	furniture	2	0	0	2	0
	fixtures	6	0	0	6	0
	structure-noise	7	0	0	0	7
	clutter/unknown	22	20	0	2	0
	<i>total</i>	57	27	2	21	7
Robot hall (det)	seating	5	4	0	1	0
	robots	3	0	1	2	0
	displays/monitors	1	0	0	0	1
	machines/electrical	1	0	0	0	1
	furniture	2	0	0	0	2
	fixtures	8	3	1	4	0
	structure-noise	8	0	0	1	7
	clutter/unknown	53	25	7	2	19
	<i>total</i>	81	32	9	10	30
Cafeteria	seating	10	1	4	5	0
	machines/electrical	1	1	0	0	0
	furniture	1	0	1	0	0
	fixtures	5	0	1	4	0
	structure-noise	3	0	0	0	3
	clutter/unknown	8	4	2	2	0
	<i>total</i>	28	6	8	11	3
T3 re-scan (owner GT)	seating	5	2	2	1	0
	robots	8	0	6	2	0
	displays/monitors	2	0	1	1	0
	machines/electrical	4	0	4	0	0
	furniture	4	0	3	1	0
	fixtures	4	0	2	2	0
	structure-noise	9	0	6	3	0
	clutter/unknown	5	4	0	0	0
	<i>total</i>	41	6	25	10	0

5.4. Owner Audit and Corrective Feedback (T3)

The preceding subsection measured what the pipeline believes; this one measures what the room’s occupant knows, and how that knowledge feeds back into the graph. It exercises the revision and upsert machinery ahead of its formal evaluation (Sec. 5.6); only the feedback mechanics matter here.

Exhaustive audit and mitigations.

The room occupant exhaustively labeled all 41 T3 clusters, giving a distinct T3 owner-audited ground-truth set beyond the showroom scoring sets. Of the 31 verified-tier clusters, 6 were correct, 13 were real objects flattened to clutter, 6 were structure-noise phantoms, 2 carried a wrong specific label, and 3 were multi-object merges (one unanswered). The absorption cause is now identifiable: the room is a robotics testbed (mobile robots, manipulators, charging stations, a disinfection robot), and none of these classes exist in the classifier vocabulary. The unverified tier behaved as designed in five structure/fixture cases but also withheld three real objects, including the wall TV of Sec. 5.3. Three audit-driven mitigations follow; their definitions, triggers, and parameters are given in Sec. 4.7 (layers 4–5), and this subsection reports their development-time evaluation. (i) *Hybrid structure filter*: registering every cluster against the exploration-scan occupancy grid — an independent, drift-free wall

prior — and combining a wall-line hug test with local (per-metre-segment) sheet thickness, a full-height band rule (z -span > 2.5 m), a ceiling band, and a room-polygon outside test removes structure noise at 12/12 recall with 3 false positives (thin wall-flush real objects: a cabinet leg, a poster stand). Crucially, the wall-flush monitor bank survives: its wall-hug fraction is 99% (the grid map draws it as wall), but its 1.8 m height and per-segment depth separate it from the 3.2 m floor-to-ceiling wall band that plane removal missed. All seven verified structure phantoms — the six structure-noise cases above plus a fluorescent fixture verified as a keyboard at 3.3 m (audited as a wrong label) — are eliminated upstream. (ii) *Recovery re-verification*: re-asking the verifier with clutter treated as *unidentified* plus a candidate shortlist recovered 4 specific types at 4/5 precision with 12 honest keeps; injecting site vocabulary (robot classes) as candidates doubled recovery (9 correct) but halved precision (8 wrong) through candidate anchoring — the TV became a “conveyor belt” when the shortlist ranked it first. (iii) *Agreement gating*: adopting a recovered label only when both passes agree, or when a pass agrees with the cluster’s own base label, yields five adoptions at 5/5 precision (two chairs, an office chair, a machine, and the TV) and resurrects no owner-refuted phantom. Domain knowledge helps recall; only agreement makes it safe. Fig. 11 shows the filter’s effect on the T3 epoch. Regression on the other epochs (no ground truth, manual inspection) flags 8/50 T1 and 18/55 T2 clusters — dominated by the documented soffit phantoms (keyboards, chairs at 2.4 m, a stair at 3.4 m) and a cluster the classifier itself had named wall — while every door, chair, machine, elevator, and wall-mounted control panel survives; the one systematic casualty is a closed-door fragment lying in the wall plane, consistent with the door-detection asymmetry of Sec. 7. Finally, refutations, corrections, and adopted labels are applied as a fourth KG revision through the same upsert history mechanism as a rescan: eight phantoms leave the graph, the poster stand is corrected in place, five recovered types land, the TV becomes its region’s key object, and the relation layer sheds nine noise-anchored edges (25 to 16 intra-place). Re-deriving ring codes and names from the revised graph heals the place layer end-to-end: the phantom-derived staircase_landing becomes display_briefing_area (TV + poster stand), the keyboard-derived workstation_area becomes machine_area, and the seating region’s recovered chairs turn cluttered_seating_area into seating_area — owner feedback propagates through all three layers of the model: object, relation, and place.

Feedback at scale: bidirectional, corrective, and time-variant.

Extending the audit loop through nine further KG revisions shows owner feedback operating in all four directions the model supports. *Restoration*: an object the owner had refuted as absent was later declared present again (the room changed between audits); a color-seeded carve at the owner-given corner recovered a 319-point red cylinder that two-pass verification typed fire_extinguisher (0.72) — the original failure was segmentation (never proposed), not classification, and owner ground truth is itself *time-variant*, which is why feedback is recorded as highest-trust evidence with history rather than as immutable truth. *Refutation*: a raw-condition ensemble find was refuted as wall paint — the patches surround the room’s wall-mounted TV, whose glossy black screen returns almost no lidar, so the “object” was the halo around a hole. *Label correction*: the correlated-bias tables of Sec. 5.2 and a poster stand (tie-break “fire extinguisher”, 0.72) were corrected in place, each with the VLM’s verdict preserved in provenance. *Attribute correction*: wall-remnant clutter was marked immovable in the implicit layer. Adjudication itself gained three inexpensive refinements along the way: canonical synonym matching (“robot_arm” vs “robot manipulator” is agreement, not disagreement), a focused tie-break pass that re-asks with exactly the two disputed labels (resolving desk vs machine at 0.62), and a coarse three-level height attribute (low/mid/high relative to the 2D-lidar scan plane and ceiling) injected as a hard physical constraint — the targeted successor to the pose-context defense of Sec. 5.8, and now a queryable implicit attribute of every node.

hybrid structure-noise filter (T3): lidar boundary-map registration $\times$ vertical-sheet geometry

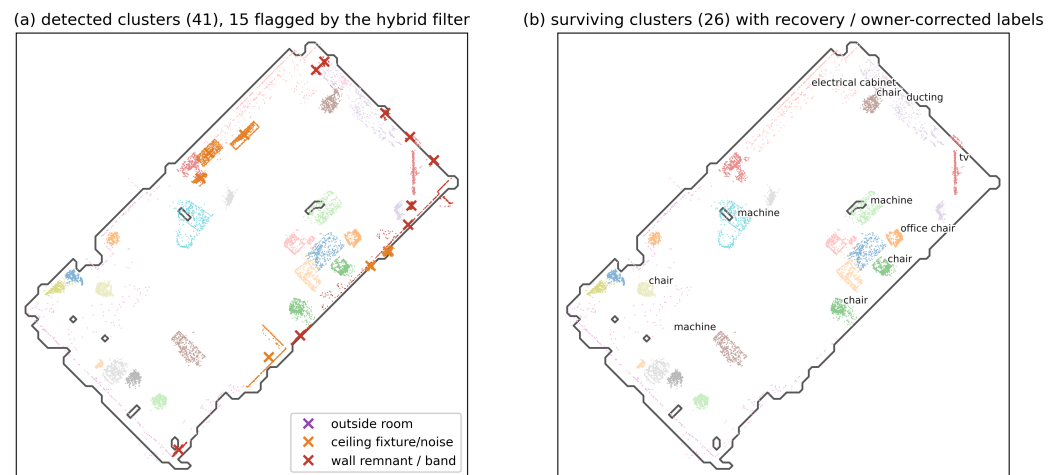


Figure 11. Hybrid structure-noise filter on the T3 epoch. (a) All 41 detected clusters; the 15 flagged ones are colored by rule — wall remnant/band (grid-map hug $\times$ thin vertical sheet, or long full-height band), ceiling fixture/noise, outside the room boundary. (b) The 26 survivors with recovery / owner-corrected labels. The wall-flush monitor bank (labeled *tv*) and wall-mounted equipment survive; residual merge labels (*electrical cabinet*, *ducting*) are retained pipeline output, not corrections.

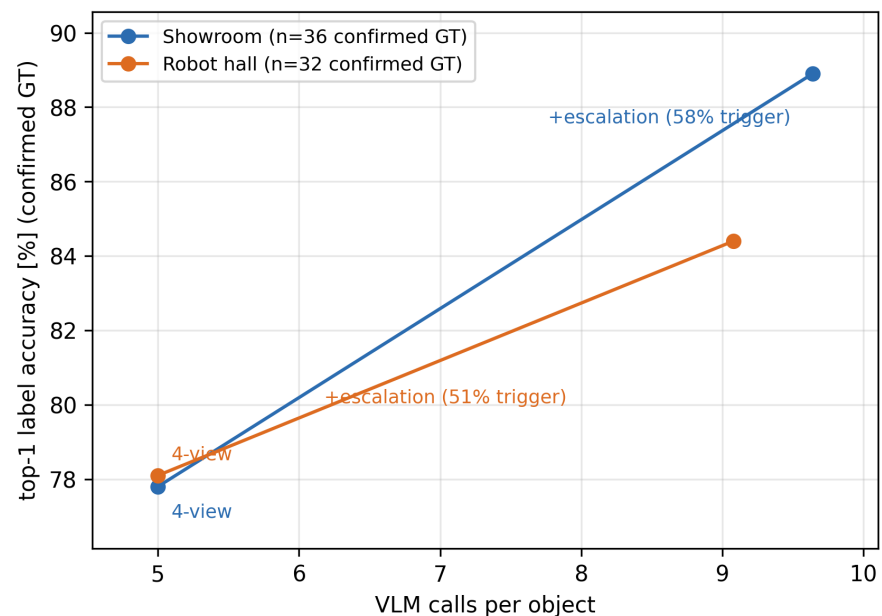


Figure 12. Escalation accuracy–cost curves for both scenes.

5.5. Knowledge-Grounded Query Evaluation

We evaluate four conditions over identically constructed per-scene query sets—48 items on the showroom, 32 on the robot hall (symbolic, explicit, scene-level, and trap questions): (1) *LLM-only* (Stage-A dump, no verification), (2) *ungated KG*, (3) *verified-only KG* (unverified nodes removed), and (4) *gated KG* (unverified retained but structurally demoted, Sec. 4.8). Each scene carries four trap questions about objects that do not exist as labeled. On the showroom all four traps target *unverified-label* phantoms (keyboard, ladder); on the robot hall two target an *unverified-label* phantom (refrigerator) and two an *escalation-verified* phantom (keyboard)—the latter probing the gate’s verifier-quality bound.

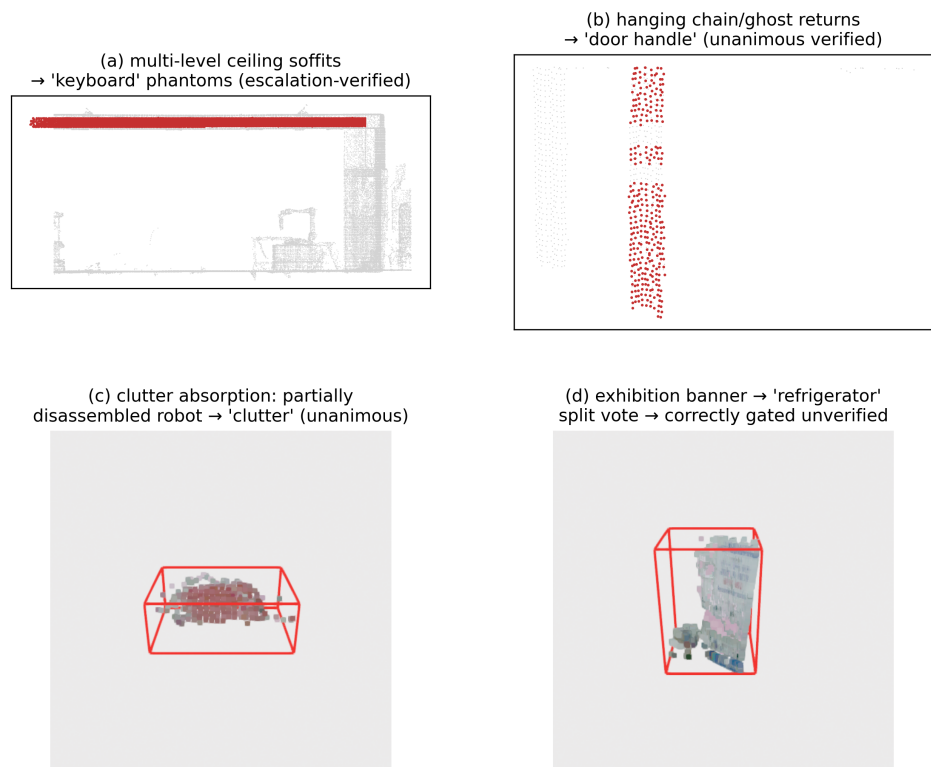


Figure 13. Verification error taxonomy: remnant phantom, unanimous-but-wrong, clutter absorption, render ambiguity.

Table 9. Query benchmark, four conditions (accuracy / hallucination rate / traps blocked), with abstention counts. Question sets: showroom $n=48$ (14 symbolic, 12 explicit, 9 implicit, 9 scene-level, 4 trap); robot hall $n=32$ (15 symbolic, 12 explicit, 1 scene-level, 4 trap). An abstention counts against accuracy (the denominator is always n) but is not a hallucination; the verified-only condition's low accuracy is largely its abstention cost.

	LLM-only	Ungated	Verified-only	Gated
Showroom ($n=48$)	52.1% / 20.8% / 2/4	81.2% / 18.8% / 0/4	58.3% / 16.7% / 4/4	83.3% / 16.7% / 3/4
abstentions	13	0	12	0
Robot hall ($n=32$)	50.0% / 37.5% / 4/4	56.2% / 43.8% / 0/4	46.9% / 40.6% / 2/4	59.4% / 40.6% / 2/4
abstentions	4	0	4	0

A correct system must decline to assert the trap labels. Note that LLM-only can “block” traps trivially: with no map it tends to deny that specific objects exist at all (4/4 on the robot hall, 2/4 on the showroom), so trap blocking is diagnostic only for the grounded conditions.

On the showroom (Table 9, Fig. 14), grounding alone (ungated) lifts accuracy from 52.1% to 81.2% but blocks no traps; removing unverified content (verified-only) blocks all four traps at a steep recall price (12 abstentions, 58.3%); the gated condition is the balance point—83.3% with 3/4 traps blocked. The two leaks are instructive and mechanistically distinct. On the showroom, one *count*-phrased trap slipped through the quarantine itself: the model tallied a demoted node by reading its `failedCandidateType` field, even though it correctly denied the same object in yes/no phrasing. On the robot hall, the keyboard phantom is *escalation-verified*, so it passes any status-based gate—gating is bounded by verifier quality; we report both as properties, not anomalies. The robot hall reproduces the ordering at lower absolute levels because its end-to-end label errors (banner→refrigerator, robot→clutter) propagate into every grounded condition—an honest end-to-end number rather than a curated one. We note the statistical scope explicitly: at $n=48/32$ questions

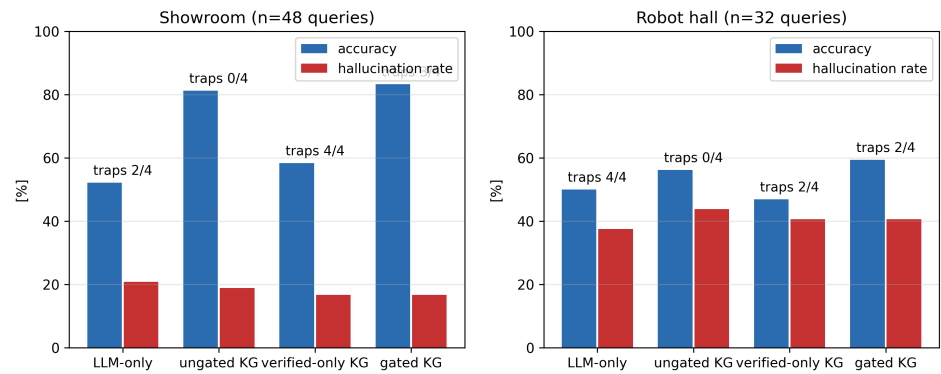


Figure 14. Four-condition query benchmark with trap-question panel.

the gated-vs-ungated *accuracy* gap is a single question on each scene, so the benchmark is diagnostic rather than powered for significance. The case for gating rests on the mechanisms the benchmark isolates—trap blocking, the abstention structure, and the direction of the hallucination change—and on the downstream mission-level consequences of Sec. 6.1, not on the accuracy deltas alone.

5.6. Incremental Update Evaluation

Synthetic scenarios.

Controlled edits of the room-scoped robot hall isolate matcher behavior (Fig. 15): (A) *unchanged repeat*: 55/55 nodes carried, zero VLM calls, zero identity switches; (B) *move*: 1/1 moved with ID preserved; (C) *remove+insert*: exact absent/insert classification; (D) *occlusion*: at 20% crop both affected objects still match (as updated, labels kept), at 40% matching fails into 2 false-absents + 2 false-news—the matcher’s quantified breaking point, and the motivation for overlap-based matching features in future work. On the combined-edit re-scan (move, insertion, and removal applied together), selective re-verification re-ran Stage B on only the two changed clusters: 26 of 822 VLM calls, \$0.23 versus \$7.37 for a full re-verification—3.1% of rebuild cost, and 0% for scenario A. We scope this claim precisely: the 3.1% figure comes from this controlled same-quality re-scan, where unchanged clusters reproduce byte-identically. On the real T2→T3 transition the scan protocol itself changed (higher-coverage acquisition), no cluster hash matched, and the epoch paid full verification cost (\$3.54 for 41 clusters)—zero-cost carry-over is a property of reproducible acquisition, not of the update mechanism alone.

Real three-epoch study.

The three epochs of Table 3 form a real update sequence: T1 (manual, June) → T2 (exploration, early July) → T3 (99.6%-coverage exploration with a controlled-change protocol, late July). Cross-epoch registration is fully automatic (FPFH [45]+RANSAC then ICP [46]; 5.1 cm RMSE for T1→T2, 4.2 cm for T3→T2). The T1→T2 diff records the room’s exhibition-to-test-hall reorganization (updated 4 / moved 3 / inserted 48 / absent 31; among the moves, a fire extinguisher displaced 3.0 m). This epoch also motivated the generic-type exclusion in the move pass: without it, clutter piles matched across 2–5 m as fictitious “moves” (9 candidates, 3 credible). The T2→T3 diff (matched 9 / moved 1 / inserted 31 / absent 45, Fig. 16) sits on top of measured natural drift: only 65% of in-room object geometry is mutually supported at 10 cm between the two scans two weeks apart—roughly a third of the room changed before any protocol was applied.

Matching scorecard. To score the real transitions without circular reference to the matcher itself, we built an independent correspondence reference by bidirectional point overlap between epochs (5 cm association, both directions ≥ 0.5 for a strong pair, 0.3–0.5 partial; the same construction as the T1 GT transfer of Sec. 5.3), and replayed the recorded upserts against it. The reference finds 4 strong and 6 partial static pairs for T1→T2 and

7 strong and 5 partial for T2→T3. Matching accuracy on strong pairs is 2/4 and 4/7; **identity switches are zero on both transitions**—every miss is a *fragmentation* error in which re-segmentation changed an extent beyond the 30% tolerance, so the object was marked absent and re-inserted under a new ID, never associated to the wrong object. The per-miss diagnosis is uniform: all five failed the per-axis extent gate (the most consequential case is the T3 “new” fire extinguisher, whose cluster overlaps a T2 clutter cluster at 0.96–0.99 mutual overlap but with width 0.11 vs. 0.28 m). Strict false-new and false-absent counts are 2/2 (T1→T2) and 3/3 (T2→T3), rising to 5/5 and 3/3 if partial pairs are also demanded; nine split/merge cases (one cluster absorbing several) are reported separately rather than forced into either bin. Multi-candidate conflicts—the greedy resolution of Sec. 4.5—occurred zero times in pass 1 on both transitions and once in pass 2, where nearest-centroid arbitration picked the owner-confirmed 0.64 m chair move. All three recorded moved claims are consistent with the reference (a true move has no in-place overlap, so the reference can refute but not confirm identity; the chair move is additionally owner-confirmed). Repeating the reference construction at 10 cm association (twice the registration RMSE) leaves identity switches at zero.

Against the controlled-change protocol, the system detected **2/2 physical changes at geometry level**: the moved chair is the epoch’s *only* moved node (0.64 m, ID preserved—a node matched continuously since T1), and the introduced floor fan was correctly inserted but typed clutter (error class (c) above), giving 1/2 at type level. Identity persistence is demonstrated across all three epochs: the fire extinguisher, pump, industrial robot, and several chairs retain their minted IDs through both reorganizations. The live graph has since accumulated thirteen revisions through the same mechanism — rescans, owner refutations/restorations/corrections, re-split decompositions, and finally a stale-node sweep in which eighteen earlier-epoch detections with no support in the current scan were marked absent; every transition, including the mistakes it corrects, is reconstructible from node histories. The place layer re-derives per-epoch names on the same SLIC regions (Sec. 4.9): exhibition-era machine_access_area and operator_control_station become clutter-dominated at T2 and machine_operator_station/electrical_services_zone at T3, at ~\$0.03 per epoch.

5.7. Retrospective End-to-End Replay against Accumulated Owner Ground Truth

After the corrective machinery of Secs. 5.2–5.6 stabilized, we re-ran the entire pipeline from the raw scan with no owner input at execution time and scored it against the live knowledge graph — by then a 46-object inventory embodying every owner restoration, refutation, and correction. Because the corrective mechanisms were developed from failures observed in this same environment, this experiment is a *retrospective end-to-end replay* on the development scene, not an independent generalization test; what it measures is how much of the owner-audited inventory the automated layers recover once owner-informed development has concluded. The replay runs the *frozen* pipeline exactly as defined in Secs. 4.1–4.7—every trigger and threshold fixed, no per-scene adjustment, no owner input at execution time—so it is an independent *run*, on a non-independent *scene*. The cafeteria (Sec. 5.8) is the only scene that is independent in both senses, and a multi-building evaluation remains future work (Sec. 7). One corrective layer transfers mechanically and can therefore be checked held-out: applying the hybrid structure filter to the cafeteria with its T3 rules frozen (the occupancy-grid prior synthesized from the scan’s floor points) flags both ceiling-noise clusters with zero false positives—both wall-flush doors survive, the trade-off that cost a poster stand on T3—while the wall pillar evades the wall-hug test and the seven carpet floor-noise clusters fall outside its rule families entirely, consistent with the floor-material gap of Sec. 7. The remaining corrective layers embed scene-specific owner evidence by construction and are not claimed to transfer. Table 10 reports cumulative geometric recall as each automated layer is enabled. The verified path alone recovers 28%; diffuseness-gated re-splitting nearly doubles it; the wall-compound gate (which self-triggers on the monitor wall and decomposes it into nine of the ten owner-confirmed units)

Controlled re-scan scenarios, room-scoped robot hall (55 objects; colored points = segmented clusters, thin boxes = explicit models)

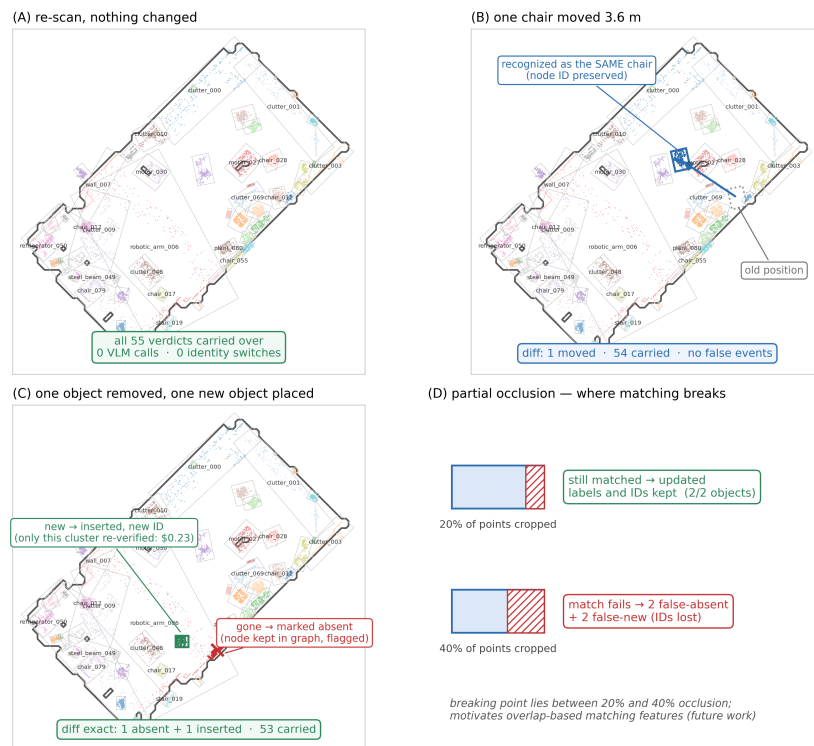


Figure 15. Controlled re-scan scenarios A–D (room-scoped robot hall, 55 objects), top-down view. Colored points are the segmented clusters, thin rotated rectangles their explicit-model footprints with node names, and the dark contour the test-room boundary; each edited object is annotated with the matcher’s decision. (A) A repeat scan with no changes carries all 55 verdicts at zero VLM cost. (B) A 3.6 m chair move is classified *moved* with its node ID preserved. (C) A removed object is marked *absent* (node kept and flagged) and a newly placed object is *inserted* under a new ID; only the insertion is re-verified. (D) Partial occlusion: both cropped objects still match at 20%, but at 40% matching fails into two false-absent plus two false-new pairs—the matcher’s quantified breaking point.

and the structure-condition ensemble bring automated recall to 85%. Type accuracy saturates far lower ($\sim 31\%$ of recalled objects) — the label bottleneck is the correlated verifier bias of Sec. 5.2, and closing it is precisely the owner layer’s role. For calibration within the lineage’s own reporting style, DK-SMF reports 78–81% detection accuracy against predefined ground-truth objects in its RGB-D environments [15]; our 85% is measured against an owner-audited inventory that includes partially scanned and wall-flush objects, though the sensing modality and ground-truth protocols differ. The re-run also reproduced the deterministic backbone’s stability end to end: the same diffuse webs, the same compound decompositions, and a first-pass verification of the fire extinguisher that had originally required owner restoration. The precision column of Table 10 tracks what recall costs: each added layer admits more candidate clusters, and the ensemble’s extra detection conditions temporarily dilute in-room precision ($60\% \rightarrow 49\%$) before the square-up step recovers part of it— automated recall is bought with review burden, which is exactly the owner-review efficiency that Sec. 5.4 argues the model should optimize.

real multi-epoch update of the same room, latest transition (rev 2 → 3)
(cross-scan registration 4.2–5.1 cm RMSE; identity carried for specific-type objects)

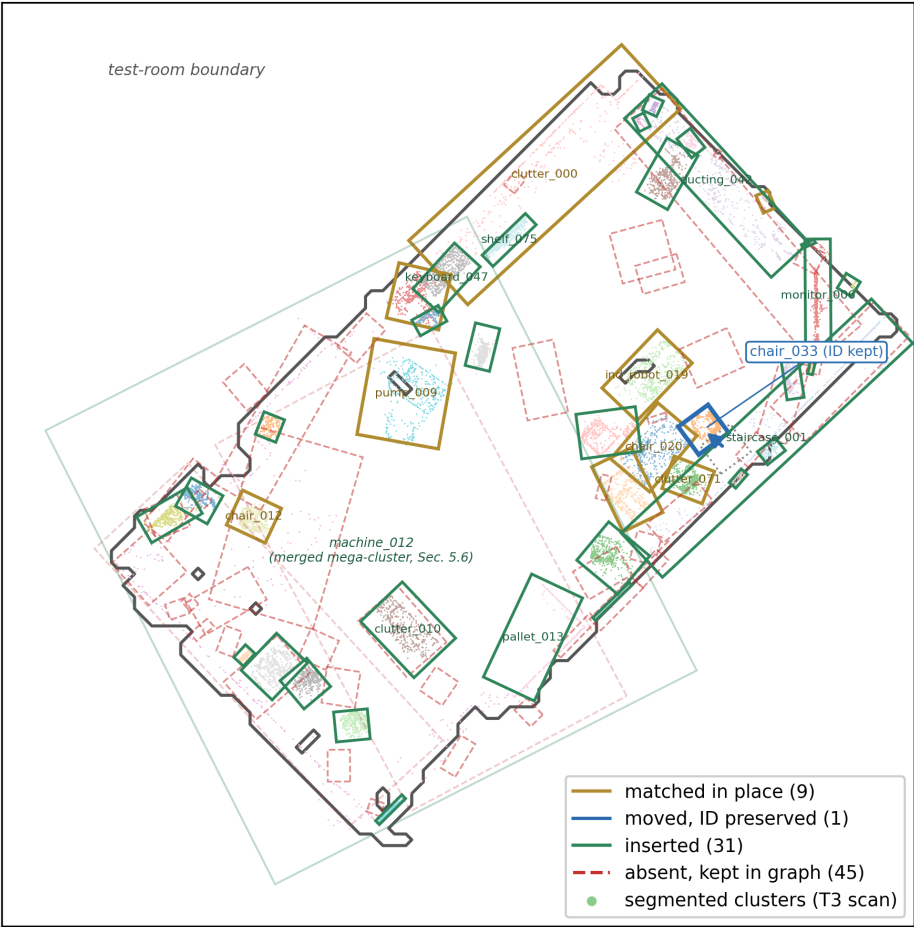


Figure 16. Real multi-epoch update, latest transition (T2→T3), top-down view. Colored points are the T3 segmented clusters inside the test-room boundary; yaw-rotated node footprints are colored by diff state (matched, moved with ID preserved, inserted, absent), with names on specific-type objects. Oversized footprints of merged mega-clusters (e.g., machine_012, Sec. 5.8) are faded.

Table 10. Cumulative automated recall of the full re-run against the owner-corrected 46-object graph (no owner input in the run itself).

Automated layers (cumulative)	GT recall	In-room precision
Verified path (Sec. 4.2–4.4)	13/46 (28%)	37%
+ diffuseness re-split	26/46 (57%)	53%
+ wall-compound gate	36/46 (78%)	60%
+ structure-condition ensemble	37/46 (80%)	49%
+ wall-compound panel square-up (Sec. 4.7, layer 2)	39/46 (85%)	51%

5.8. Ablation and Sensitivity

Escalation on/off.

Disabling escalation reduces the pipeline to its base-4 fusion, already tabulated as the base-4 column of Table 6: escalation contributes +11.1 points on the showroom and +6.3 on the robot hall, at the bounded call cost analyzed in Fig. 12.

Gating style.

Instructing the model to respect verification status blocks 1/4 traps; structural demolition blocks 4/4 under identical prompts in this isolated comparison (vs. 3/4 end-to-end,

Sec. 5.5)—still the single strongest argument that reliability must live in the data contract, not in instructions.

Pose context.

Injecting mounting-height context as a targeted defense removed all three phantom ceiling keyboards in its pilot, but applying it to every object *reduced* escalated accuracy (88.9%→75.0%) and escalated precision (100%→72.7%)—a desk-height prior that helps ceilings hurts everything else, so it is deployed as a targeted defense only.

Uni3D vocabulary.

Swapping the Stage-A classifier vocabulary ranks generic-30 (66.7%) above curated-20 (48.7%) and above the owner-derived 18-term set (33.3%), with ~5-point run-to-run variance—narrowing the prompt set toward the deployment vocabulary *hurts* the embedding-level classifier, confirming the domain gap that motivates VLM verification downstream.

Provisional-label ablation.

Since the Stage-A label is supplied to the verifier as context, a fair question is whether Stage B verifies or merely echoes it. We re-ran the full showroom protocol (same renders, same verifier, base + automatic escalation) with the provisional label removed from both the system prompt and the per-object context—dimensions and all perceptual guardrails retained. The answer has two sides. With the label, Stage B is a genuine but conservative corrector: on confirmed GT it departs from Stage A on 8 of 36 objects, 5 correctly and 3 incorrectly, turning Stage A's 77.8% into 88.9%. Without the label, the verifier is not an independent classifier: decisiveness collapses (trigger rate 58% → 90%, unanimity 21 → 5 objects) and escalated accuracy falls to 38.9%, with 22 of 26 departures from the Stage-A label wrong. Two mechanisms account for the drop. Generic piles lose their anchor: blind, the verifier names a specific type for clutter (12 vs. 23 clutter answers against 22 GT), and on specific-type objects accuracy still halves (10/14 → 4/14)—sparse renders under-determine the label, and the prior selects among render-compatible hypotheses. The tier machinery, however, degrades in the intended direction: blind uncertainty flows to unverified (38 vs. 21 objects) rather than into confident errors, though verified-tier precision does erode (unanimous 0.667, escalated 0.5). The conclusion we draw is the one the architecture already assumes: Stage B is a *verifier of hypotheses*, and the provisional label is load-bearing context—the same anchoring power that the candidate-shortlist experiment of Sec. 5.4 shows can mislead, which is why label adoption is agreement-gated and owner-auditable rather than trusted outright.

Verifier model.

Running the identical Stage-B protocol (Table 11) across five verifiers (two vendors, three tiers, two generations; reasoning disabled for parity) makes escalation an economic instrument — call volume tracks verifier decisiveness monotonically (247 to 383 calls as unanimous verifications fall from 26 to 9). Three findings. First, within a generation the smaller tier splits more votes and resolves fewer: its indecision inflates call volume by 14% (383 vs. 335 calls), clawing back part of its nominal 3× per-token price advantage — and what survives is paid for in the lowest type accuracy and the largest unverified residue (17 of 35 objects). Second, all three current-generation frontier models tie on type accuracy against the owner inventory (7/13), but differ two-fold in the calls they need to get there — the same-generation mid-tier (Sonnet 5) is the most decisive (26 unanimous, 9 escalations), and the premium tier's extra deliberation (18 escalations) buys no additional accuracy here. Third, the ranking is generation-dominated, not vendor-dominated: both 2026-generation mid-tiers beat both older Claude tiers, and cross-vendor differences within the generation are smaller than the tier effects. Escalation volume is thus a label-free, online proxy for verifier decisiveness on the deployment's own data — and a practical criterion for deployment-time verifier selection under *cost* constraints: pick the cheapest verifier whose

Table 11. Verifier-model comparison: identical Stage-B late fusion + escalation on the same 35 in-room clusters, scored against the owner-corrected graph.

Verifier	Verified	Esc.-resolved	Unverified	Type-correct ¹	Calls	Cost ²
Claude Sonnet 5	26	3	6	7/13	247	\$2.49
GPT-5.6 (luna)	21	7	7	7/13	287	\$0.11
Claude Opus 5	17	8	10	7/13	319	\$5.28
Claude Sonnet 4.6	15	6	14	6/13	335	\$3.00
Claude Haiku 4.5	9	9	17	5/13	383	\$1.13

¹ canonical-synonym type match on the clusters that geometrically match a ground-truth object. ² measured tokens (in/out: 0.63M/40K, 0.42M/19K, 0.79M/53K, 0.74M/52K, 0.84M/59K, row order) at vendor list rates of August 2026 (\$/Mtok in/out: 3/15, 0.20/1.20, 5/25, 3/15, 1/5). The GPT-5.6 row reflects the 2026-07-30 price reduction; at its launch rate (1/6) the same run costs \$0.54. Sonnet 5's introductory rate (2/10) gives \$1.66.

Table 12. Cafeteria stress scene: verification under verifier $\times$ render source (the identical 512-px multi-view protocol of Sec. 4.4 throughout). Accuracy is escalated accuracy on confirmed GT ($n=20$; the fragment-excluded column drops the five confirmed sofa-fragment clusters, $n=15$); unanimous precision is over confirmed unanimous votes. In every cell, 0 of the 7 sofa clusters are recovered.

Verifier	Render source	Esc. acc.	(excl. fragments)	Unan. prec.	Trigger
Sonnet 4-6	3 cm voxel cloud	35.0%	46.7%	57.1%	68%
Sonnet 4-6	full-resolution	40.0%	53.3%	62.5%	60%
Sonnet 5	3 cm voxel cloud	45.0%	60.0%	66.7%	40%
Sonnet 5	full-resolution	60.0%	80.0%	80.0%	52%

escalation rate has saturated. It is not a proxy for accuracy: the upgrade check below shows the most decisive model trading unanimous precision for its decisiveness on the two full GT scenes, so quality must still be priced by a small labeled probe set or by accumulated owner feedback. The cost guide is robust where prices are not: a vendor repricing days before our runs moved one verifier's end-to-end cost five-fold (Table 11, note 2), while the decisiveness ordering it would be applied to is measured on the deployment's own data.

Verifier upgrade check.

Because the ground truth is fixed and scoring is automatic, the headline study of Sec. 5.3 re-runs unchanged under a different verifier. Re-running both scenes under Claude Sonnet 5—the most decisive model above—confirms the decisiveness transfer: escalation triggers fall from 58% to 50% (showroom) and from 51% to 33% (robot hall), unanimous verifications rise from 21 to 25 and from 25 to 34, and escalation calls drop from 232 to 200 and from 208 to 136. The freed decisiveness is not free accuracy, however: unanimous precision falls from 100% to 91.7% and from 94.4% to 83.3%, and escalated accuracy lands at 69.4% and 81.2% (vs. 88.9% and 84.4%)—showroom deltas several times the ± 3 -point run-to-run variance measured in Sec. 5.3, and directionally reproduced on the second scene. The dominant new failure is clutter absorption—real chairs, junction boxes, and a control panel unanimously verified as clutter—i.e., the newer model re-distributes error mass from indecision into confident under-labeling on these 512-px render sets, while on the 1024-px renders of Table 11 it ties for the best type accuracy. Verifier upgrades therefore change *where* errors live rather than uniformly removing them, which is precisely why the framework treats verification as tiered evidence behind a gate rather than as truth: the tier structure, the escalation economics, and the owner loop survive a verifier swap; the specific precision figures do not.

Cross-floor stress scene.

The cafeteria scene (Sec. 5.1) probes the same protocol on a furnishing domain the campaign never saw: carpeted floor, fabric sofa booths, planter partitions (Fig. 17a). The backbone transfers mechanically—57M points to 28 clusters in 13.5 s after an automatic

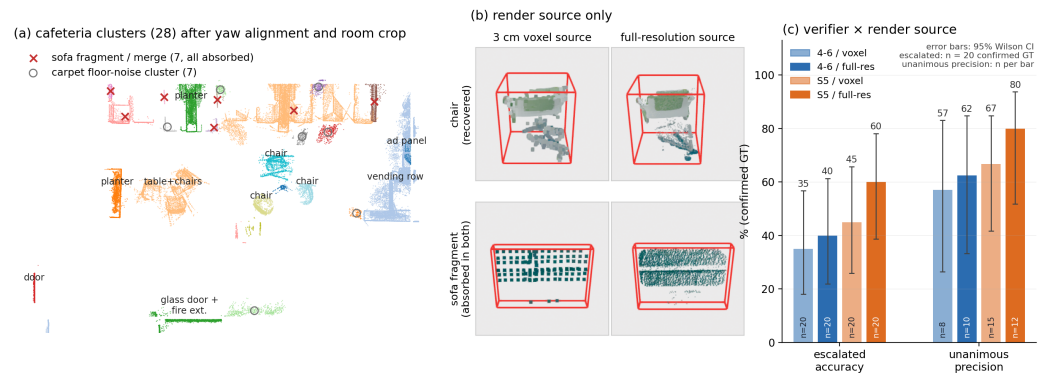


Figure 17. Cafeteria stress scene. (a) The 28 clusters after automatic yaw alignment and room crop; crosses mark the seven sofa fragments/merges (absorbed as clutter in every condition of Table 12), circles the seven carpet floor-noise clusters. (b) One cluster, rendered from the 3 cm working cloud (left) and from full-resolution re-associated points (right) under the identical 512-px protocol: the chair (top) is recovered by the render-source change alone; the sofa fragment (bottom) is not. (c) Escalated accuracy (confirmed GT, $n=20$ per cell) and unanimous precision (per-bar $n = 8\text{--}15$ unanimous objects, annotated on each bar) across verifier $\times$ render source; error bars are 95% Wilson intervals, wide on the precision bars because unanimous sets are small—the point estimates order the cells, the intervals bound how far.

wall-normal yaw alignment—but verification does not (Table 12, first row): escalated accuracy reaches only 35% against confirmed owner GT, and unanimous precision falls to 57.1%, with two unanimous keyboard phantoms on carpet-noise clusters and a chair unanimously absorbed as clutter (a fourth unanimous miss, a table pedestal, falls on plausible-graded GT). The errors are structured, not diffuse. The scene’s dominant object family—the sofas—enters the pipeline pre-broken, split into five cushion fragments and two sofa–table merges, and every one is absorbed (crosses in Fig. 17a); carpet pile leaves seven residual floor-noise clusters above the RANSAC floor plane (circles; the floor-material analogue of the ceiling remnants of Sec. 5.2); and the confidence gate quarantines 9 of the 16 wrong labels over the 25 scored clusters as unverified while the unanimous errors pass.

Render source $\times$ verifier.

Two upgrades are then applied factorially (Table 12, Fig. 17c): re-rendering the identical 512-px view sets from full-resolution re-associated points instead of the 3 cm working cloud (isolating render quality at unchanged protocol and cost, $\sim \$2.5/\text{scene}$), and swapping the verifier for Sonnet 5. The two levers act on different error classes. The render-source change repairs *whole-object* ambiguity for both verifiers: under Sonnet 4-6 the keyboard phantoms vanish, both chairs recover (including the unanimous miss; Fig. 17b, top), and both doors resolve correctly, one via escalation—while the voxel renders block the same chairs and doors under Sonnet 5 as well. The verifier upgrade adds accuracy on top of whichever source it is given (35 $\rightarrow$ 45% on voxel, 40 $\rightarrow$ 60% on full-resolution sets): on this scene the newer model is uniformly better—in contrast to its unanimous-precision loss on the industrial 512-px sets in the upgrade check above—so even the *sign* of a model upgrade is domain-dependent. The render-source lever carries the same caveat: symmetric controls that re-render *both industrial scenes* from full-resolution points reverse the effect—showroom escalated accuracy drops from 88.9% to 80.6% (base 77.8% $\rightarrow$ 69.4%; unanimous precision holds at 100%) and robot-hall from 84.4% to 68.8% (unanimous precision 94.4% $\rightarrow$ 83.3%). The added texture invites over-specific labels—a junction box read as a pipe assembly, the wall TV as a ladder, clutter piles as keyboards, boxes, and trash cans—where the coarser working-cloud render had left a correct verdict, so the main campaign’s voxel-source protocol stands as the right default for the industrial scenes, and the cross-scene comparison (first row of Table 12 against Table 6) is same-protocol by construction. What

neither lever reaches is fragmentation: the seven sofa clusters are absorbed in all four cells (Fig. 17b, bottom) and the planters stay mislabeled. Segmentation bounds what any verifier can see; render quality bounds what a given verifier resolves; the model generation buys accuracy only on top of both. Together with the upgrade check, the stress scene fixes the reading of Table 6: tier precision is a scene- and domain-scoped measurement, to be re-estimated per deployment—cheaply, via a small owner-confirmed probe set—rather than a property of the architecture.

Segmentation parameters.

The backbone’s parameter sensitivity (36-combination sweep) and the conditional refinement rules are ablated where their context lives, in Sec. 5.2.

Voxel resolution and registration noise.

Two controlled perturbations of the robot-hall backbone, scored per object against the owner-grounded reference of Sec. 5.2, bound the geometric operating regime. Voxel size is not a free knob but part of a jointly calibrated triple with $(\epsilon, \text{min_points})$: at the baseline 3 cm the full backbone recovers 42/42 reference objects, but refining to 2 cm collapses recall to 0.36 (density rises, the fixed $\epsilon=0.30$ bridges gaps, 22 objects merge) and coarsening to 5 cm collapses it to 0.33 (small objects starve below the 100-point floor, 18 misses)—so the voxel size must be reported with the clustering parameters, as Table 4’s pipeline configuration does. Seeded per-point Gaussian noise applied to the raw cloud before the backbone probes registration error at the point level: at $\sigma=5$ mm—the scale of the scanner’s real per-scan bundle error (4–5 mm)—recall drops to 0.48, and by $\sigma=20$ mm ($4\times$ bundle error) the backbone effectively fails (0.07), the dominant mechanism being saturation of the 5 cm RANSAC inlier band: un-removed wall residue grows from 267K to 853K points and chains objects into merged clusters (39/42 merged) rather than fragmenting them. We note the regime carefully: per-point scatter at these levels is an artificial stress—the real ~ 5 cm inter-epoch ICP residual is a rigid-transform error the pipeline absorbs at the KG matching level (Sec. 5.6), not per-point noise—so the practical claim is robustness at the real bundle-error scale, contingent on the structure-removal stage, which is the first component to saturate. Color variation is not isolated by any experiment in this paper: the render-source ablation above changes point density and silhouette detail together with color, so it is an appearance-quality control rather than a pure photometric one, and a color-only perturbation study remains open.

6. Application Demonstrations

This section demonstrates the framework in use. The semantic mission experiment of Sec. 6.1 is quantitative; the twin injection of Sec. 6.2 illustrates the single-store integration surface; and Sec. 6.3 repeats the mission interface on the physical AMMR, with the console, the twin, and the robot all served from the same knowledge graph.

6.1. Semantic Mission Execution in Simulation

We close the loop from scan-derived knowledge to autonomous behavior. The TOSM robot layer is instantiated from the platform’s URDF and navigation configuration (footprint, drive type, sensor mounts, velocity limits) and ingested alongside objects and places, completing the three-element TOSM model. A *semantic mediator* resolves natural mission commands — `goto_place`, `goto_object`, `patrol`, `return_home` — against the knowledge graph: places resolve to free-space points at region centroids, objects to approach poses computed from the target’s footprint and the robot’s radius—the kinematics-aware resolution style of TOSMNav [14], which checks robot footprints against doorway entrance sizes in its planning rules—with the verification gate deciding eligibility. The mediator covers command resolution only; full hierarchical task planning over the same TOSM knowledge (PDDL/POPF in [14]) is the natural next consumer of the graph and is out of scope here. A hybrid deliberative/reactive behavior tree [47,48] (Fig. 18) executes the resolved goals

Table 13. Phantom-goal mission waste in simulation (AMMR proxy). The same three commands are resolved against the knowledge graph before and after structure filtering + owner feedback (both with the verification gate active). Wasted path includes the return leg.

KG state	Phantom target (truth)	Outcome	Wasted path	Wasted time
pre-feedback	keyboard (fluorescent lamp)	driven	1.7 m	16.8 s
pre-feedback	staircase (bare wall)	false arrival	0.6 m	5.7 s
pre-feedback	book (wall pillar)	unreachable, aborted	0 m	22.2 s
filtered + owner feedback	all three	refused	0 m	~1 s

through Nav2 [41], with a reactive layer that preempts the mission on low battery or an interrupt and returns the robot to its home pose. The management console (Sec. 3) issues commands and edits the graph live; the mediator hot-reloads owner edits, so refuting a phantom in the UI changes robot behavior on the next command without a restart.

Experiments run in a Gazebo [49] world generated from the same exploration occupancy grid that anchors the place layer, so world, navigation map, and knowledge graph share one frame. The simulated vehicle is a digital proxy of the target platform itself: the AMMR’s footprint (1.25×0.8 m), lidar mount, and velocity limits from the robot layer are instantiated in the world, and the mediator derives approach poses and clearance checks from the same layer (circumscribed radius 0.74 m) — the knowledge that *describes* the robot is the knowledge that *drives* it. Mission-level metrics: over three repetitions of a three-mission battery (approach the TV, go to the seating area, return home), task success is 9/9, with mean leg path lengths 4.6/5.4/1.8 m and completion times 16.5/24.9/15.0 s; semantic resolution latency (command to accepted or refused goal) is under one second, and a gate refusal stops the robot safely with zero motion. The reactive layer supplies the safety-stop behavior: a low-battery event during navigation preempts the active Nav2 goal and returns the platform to a holding position at home, resuming only on recovery. The phantom experiment (Table 13) sends the robot to three owner-refuted phantoms under two knowledge states. Against the pre-feedback graph, all three phantoms carry *verified* labels, so the confidence gate passes them — and the waste takes three distinct forms: a pointless drive to a fluorescent lamp; a *false arrival* at a 7.4 m “staircase” whose approach ring encloses the robot’s own position, so the mission reports success at a structure that does not exist; and 22 s of futile replanning toward a wall-pillar “book” whose approach pose the platform’s footprint cannot reach at all. Against the graph after structure filtering and owner feedback, the same commands are refused in about one second with zero motion. Verification gating alone cannot stop verified phantoms; it is the combination of gating, structure-noise removal, and owner feedback that reduces the robot’s exposure to unverified and phantom semantic claims. Fig. 19 shows a gated mission trajectory and an ungated phantom drive.

6.2. Simulation Twin Injection

Because every node carries an explicit model in the scan frame, the graph injects directly into a physics simulation. We reuse the digital-twin construction of our prior work [50] and add the semantic layer: each verified object spawns a labeled proxy (extent box or segmented cloud) in NVIDIA Isaac Sim, with verification status, provenance, height level, and label stability exposed in the scene inspector. Unverified objects spawn as neutral gray placeholders—the gate is visible in the twin. The knowledge graph is the *single* semantic store: the management console and the mission mediator read the same file live, and the Isaac stage is re-derived from it (in the shared map frame) whenever the graph changes, so an owner edit propagates to the robot’s next goal, the console, and the twin from one write. The discipline is self-auditing — deriving the stage from the graph immediately *visualized* eighteen stale earlier-epoch nodes as orphan boxes that per-view export paths had silently hidden, prompting the cleanup revision of Sec. 5.6.

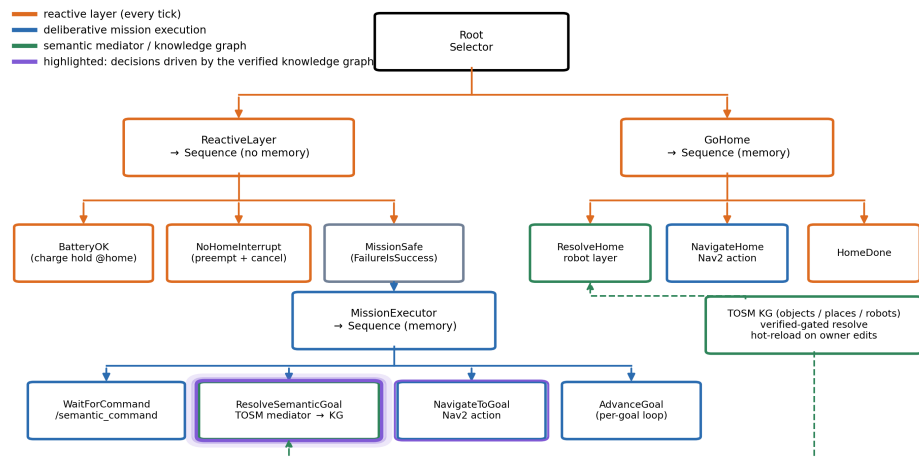


Figure 18. Hybrid deliberative/reactive mission behavior tree. The reactive layer re-evaluates battery and interrupt conditions every tick and can preempt navigation (canceling the active Nav2 goal); the deliberative branch resolves semantic commands through the TOSM mediator, which hot-reloads the knowledge graph when the management console edits it. Highlighted nodes (violet) are the decisions the verified knowledge graph drives: the semantic-resolve node, where a gated graph refuses phantom or unverified goals, and the navigation node that consumes the resolved pose; the unhighlighted reactive nodes are generic behavior-tree machinery.

6.3. Real-Robot Mission Execution

To close the loop on hardware, the mission stack of Sec. 6.1 was run unchanged against the physical AMMR in the test room (the robot hall configuration), with the knowledge graph of Sec. 5.6 as the semantic store. The platform runs its own localization and Nav2 stack on an independently built occupancy map; a thin bridge node replaces the simulator: it converts the robot's localization pose into the scan frame through a single SE(2) offset, calibrated once by parking the robot on a taped start spot of known scan coordinates, and forwards the mediator's resolved goals (scan frame) to the robot's on-board navigation in the robot's map frame. Nothing above the bridge changes: commands are issued from the management console, the mediator resolves and gates them against the graph, the behavior tree executes them, and the Isaac Sim twin mirrors the robot from the same store. Fig. 20 shows two filmed missions and Table 14 summarizes all runs; Video S1 contains the synchronized console, twin, and camera footage.

Three semantic drives completed — two goto_object approaches to the wall TV and one goto_place drive into the monitor workstation region, each 3.6–4.0 m from the start — in 14–22 s with arrival errors of 0.22–0.34 m (mean 0.30 m) between the robot's localized pose and the resolved goal; a place command issued while the robot already stood inside the region completed without motion (0.11 m from the region goal). The gate behaved as in simulation: goto_object keyboard, the owner-refuted phantom of Table 13, was refused by the mediator in under one second with zero robot motion, twice. The run also exposed a platform-side limit that the semantic layer cannot mask: after a completed mission the robot's navigation stack accepted but did not execute a subsequent goal in two of two attempts (a 1 m direct goal was not executed either), while the bridge log shows the mediator's resolution and dispatch as correct in every case; missions were therefore run one per platform initialization, and we report this as a property of the platform's navigation stack, not of the framework. The exercise is a bounded validation — one room, one platform, three completed drives — but it confirms that the graph that survives verification, gating, and owner feedback is directly consumable by a real robot, and that the twin, console, and robot stay consistent because they read one store.

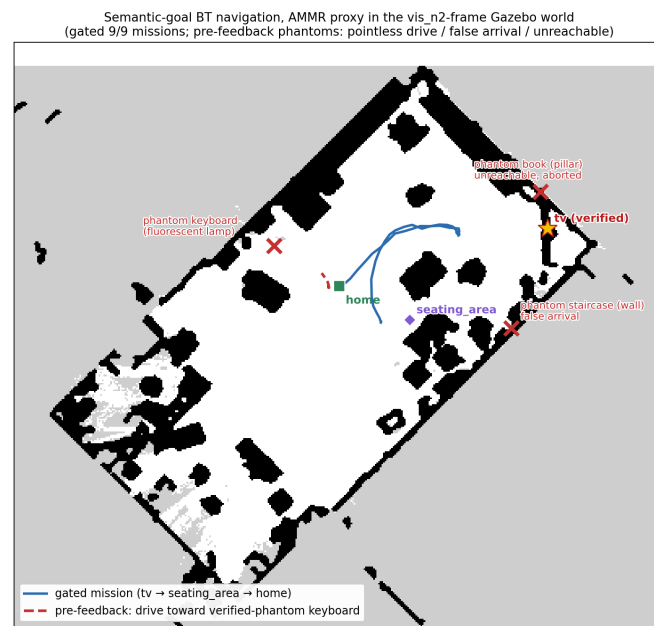


Figure 19. Semantic-goal navigation of the AMMR proxy in the Gazebo twin of the test room (world, map, and knowledge graph share the exploration-scan frame). Blue: a gated three-mission run (TV → seating area → home). Red dashed: against the pre-feedback graph the robot drives toward a verified-phantom keyboard — a fluorescent lamp; crosses mark the three phantom goals with their failure modes (pointless drive, false arrival, unreachable). All three are refused once structure filtering and owner feedback are applied. These runs establish the knowledge-graph-to-mission interface in a physics simulation; Sec. 6.3 repeats the interface on the physical platform.

Table 14. Real-robot semantic missions on the AMMR (test room, 15 Sep 2026). Goals are resolved by the mediator in the scan frame; distance is straight-line start-to-goal; arrival error is the localized-pose-to-goal distance at mission completion.

Command	Resolved goal (m)	Dist.	Time	Err.	Outcome
goto_object tv	tv_006 approach (3.09, 1.91)	4.0 m	14 s	0.34 m	completed
goto_object tv (Fig. 20a)	tv_006 approach (3.09, 1.91)	3.6 m	20.8 s	0.33 m	completed
goto_place electrical_maintenance_area (Fig. 20b)	region (−0.69, 3.81)	3.9 m	22.1 s	0.22 m	completed
goto_place display_briefing_area	region (3.31, 1.76)	0.1 m	<1 s	0.11 m	completed (already inside)
goto_object keyboard (×2, phantom)	—	—	<1 s	—	refused by gate, no motion
2nd goal after a completed mission (×2)	resolved, dispatched	—	—	—	not executed by platform; re-init

7. Discussion

Cost and latency.

VLM verification is the only paid stage. Escalation is bounded by its trigger rate (51–58% on the main-campaign scenes; 68% on the stress scene), and per-call cost is ~\$0.009; full verification averages ~\$0.09 per object (\$3.63 for a 38-cluster run and \$7.37 for the 86-cluster robot-hall det run—the full 88-cluster detection set before room scoping and remnant filtering, of which 86 had renderable view sets—all calls included). The incremental design changes the cost model qualitatively: an identical-input repeat scan costs \$0 (hash carry-over) and under reproducible acquisition a changed scene can cost approximately in proportion to its detected change (3.1% of rebuild in our controlled same-quality re-scan; a transition that changes acquisition quality, like the real T2→T3, forfeits the carry-over and pays in full, Sec. 5.6). Wall-clock time is equally bounded, measured on this workstation: the full deterministic backbone—loading the 68.6M-point E57, 3 cm downsampling, ten-plane removal, decomposition, and remnant filtering—completes in 29 s (8.1 GB peak memory), rendering the four-view sets takes 99 s for 88 objects, and verification with escalation runs in 7.7 min for a 50-object scene at six concurrent calls, so the complete semantic modeling

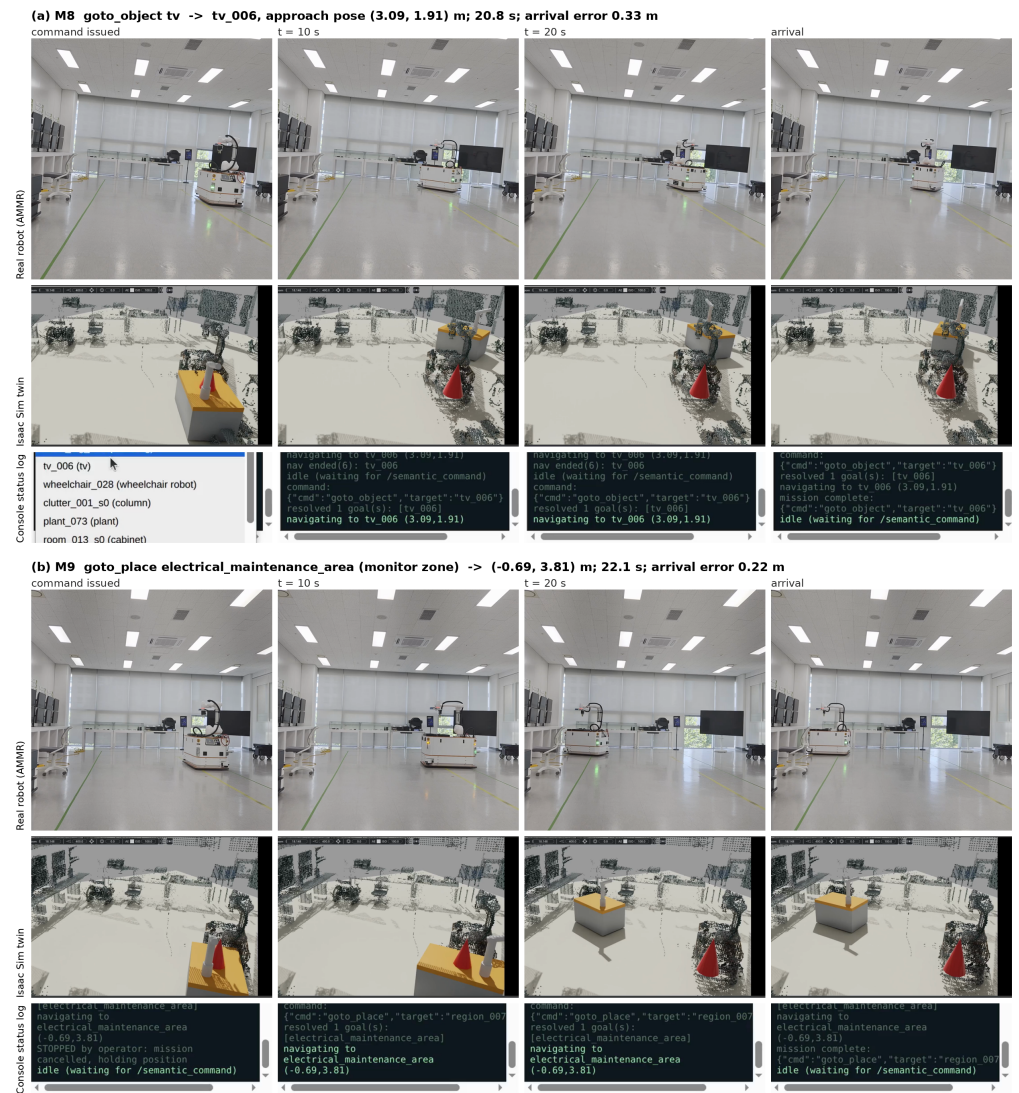


Figure 20. Real-robot semantic missions (Table 14). Rows: the physical AMMR, the Isaac Sim twin derived from the same knowledge graph (the red cone marks the twin’s goal handle), and the management console’s status log; columns: command issued, 10 s, 20 s, and arrival. (a) goto_object tv: the mediator resolves the verified node tv_006 to an approach pose and the robot arrives 0.33 m from it in 20.8 s. (b) goto_place electrical_maintenance_area: the region containing the monitor workstation, reached in 22.1 s with 0.22 m error. Video S1 shows the synchronized footage.

of a scene finishes in well under ten minutes end to end. Verification is offline; no VLM call sits on a robot’s control path.

Residual unverified content.

On exploration-quality scans, 24% of objects remain unverified (47% on the manual scan; 43% on the 1/6-coverage ablation)—scan quality is the dominant driver of verification yield. The framework’s position is to expose this residue, not shrink it cosmetically: unverified nodes stay queryable with demoted types, and the gate is only as good as the verifier (Sec. 5.3’s unanimous-but-wrong cases; the escalation-verified keyboard trap that survives gating on the robot hall). Structural gating prevents unsupported use of *unverified* labels but cannot compensate for false-positive verification; verifier-bounded gating is a property users must design around, and we report it as such.

Small low-reflectance objects.

The clutter-absorption error class—disassembled robot, poster stand, and the controlled-change floor fan all verified as `clutter`—shows a systematic blind spot: geometrically detected, semantically flattened. Change detection still works (the fan was correctly inserted), but type-level recall for small dark appliances needs either close-range imagery or targeted re-rendering. The cafeteria render-source ablation (Sec. 5.8) tests exactly this mechanism and splits it: re-rendering from full-resolution points repairs the *whole-object* errors it is aimed at (keyboard phantoms, chairs, doors), while absorption of object *fragments* survives untouched. The industrial-scene controls bound the lever from the other side—the same upgrade *lowers* accuracy on both industrial scenes by inviting over-specific labels—so targeted re-rendering is the right response only when segmentation has delivered the object intact *and* the working-cloud render is what destroys its identity. The T3 owner audit (Sec. 5.4) adds a vocabulary dimension: when the deployment site contains object families absent from the classifier vocabulary (robots, in our testbed), absorption becomes the dominant error, and naively injecting site vocabulary as candidates trades precision for recall through candidate anchoring; agreement gating restores precision at reduced recall.

Structure remnants.

The soffit filter removes vertical transition bands, but horizontal and sloped ceiling patches and content seen through openings survive structure removal (six residual cases in the robot hall). These are bounded, enumerable artifacts. The T3 audit shows the practical fix: a hybrid filter that registers clusters against the exploration-scan occupancy grid recovers all remaining wall, ceiling, and outside-boundary remnants (12/12) while keeping wall-flush furniture, at the cost of thin wall-flush objects (a poster stand) — geometry alone cannot separate a 0.1 m sheet standing at the wall from the wall itself. Structure removal has a second, opposite failure mode that the ensemble of Sec. 5.2 exposes: it *amputates* wall-flush objects (two multi-metre tables, a wall-mounted monitor bank) along with the wall. Both failure directions argue for treating removal as a varied condition to detect across, not a fixed preprocessing truth. The cafeteria adds a floor-material variant: carpet's rough, low-reflectance pile leaves residual patches above the RANSAC floor plane, and seven of that scene's 28 clusters are such floor noise—the smooth-plane assumption behind structure removal is itself a material assumption.

Correlated verifier bias.

Cross-condition voting corrected no labels: when the verifier mislabels, it tends to mislabel the same way regardless of how the input was segmented (the correlated conveyor-belt case of Sec. 5.2). Ensembling over *inputs* cannot decorrelate a bias that lives in the *model*; what the ensemble does yield is an empirical instability signal (`labelStability`) that tells the owner where to look. Together with the time-variant ground truth of Sec. 5.4, this shapes our position: the owner is not a fallback annotator but the highest-trust evidence source in a permanently open loop, and the model's job is to make each owner minute maximally productive — surfacing the unstable, preserving provenance for every overridden verdict, and propagating one correction through object, relation, place, scene, and mission layers at once.

The coverage paradox.

Better scans can make density-based decomposition worse: at 99.6% coverage, gaps between adjacent objects close and DBSCAN chains 41% of the scene into one mega-cluster. Our measurements show this artifact accounts for only ~12% of the epoch's absent nodes (of the 43 re-associable absents, 5 recoverable by tighter re-splitting and 4 by Point-SAM refinement)—the rest is genuine change—but the effect argues for coverage-adaptive clustering or learned instance priors as the backbone's next iteration, applied under the same flagged-candidate discipline as our Point-SAM rule.

Scope.

Results come from one scanner, one building, and object-level change; structural change (walls, large fixtures) and cross-site generalization are untested. The cafeteria stress scene extends coverage to a second floor and a soft-furnishing domain, but as a bound rather than a validation: it shows the headline precision figures do not transfer to out-of-vocabulary furnishing without segmentation-level and render-level adaptation (Sec. 5.8). Manipulation is likewise out of scope: the implicit layer plus verified pose and extents provide the minimum contract a manipulation planner needs for coarse feasibility screening, but we state this as an interface property without experiments. The three-epoch study mitigates but does not remove the two-scene limitation. Real-robot validation (Sec. 6.3) is bounded in the same way: three completed drives and two gate refusals in one room on one platform, with the platform re-initialized between missions because its navigation stack did not execute consecutive goals; the semantic layer's resolution and dispatch were logged as correct throughout, but a multi-mission, multi-site campaign remains to be done. The place layer's regions are parameterized SLIC superpixels over the grid map: spatial cells rather than architecturally meaningful rooms, with the superpixel count fixed by hand; inferring wall- and doorway-aligned region boundaries is future work.

8. Conclusion

We presented TLS-SMF, a framework that turns registered terrestrial laser scans into a queryable, confidence-aware, and maintainable semantic knowledge graph for indoor robots. A deterministic geometric backbone makes decomposition reproducible and diffs meaningful; multi-view VLM verification with automatic escalation attaches empirically validated confidence tiers to every label while retaining what it cannot resolve as explicitly unverified; and mint-once identities with two-pass matching turn re-scans into node-level upserts whose verification cost can scale with detected content change under reproducible acquisition. Across two operational configurations of one industrial room, a real three-epoch study of that room, and a cross-floor stress scene that bounds domain transfer, the framework improved verification accuracy by up to 11 points through escalation, blocked hallucination traps by gating the data rather than instructing the model, preserved node identities for continuously tracked objects across three scan epochs—with zero observed identity switches on strong reference pairs, though re-segmentation still yields false-new/false-absent cases—including a controlled physical change, and re-verified identical-input repeat scans at zero marginal VLM cost. Extending the loop beyond one-shot verification, automatic re-split gates and a structure-condition ensemble raised automated recall against an owner-audited 46-object inventory—owner-free at execution time, after owner-informed corrective development (Sec. 5.7)—from 28% to 85% — including a nine-of-ten decomposition of a monitor wall the baseline had verified as a single cabinet — while the remaining type errors localized the owner's role precisely: verifier bias is correlated across models, conditions, and view sets, so label truth enters through revisioned owner evidence over a single semantic store that also feeds the console, the mission mediator, the simulation twin, and—through the same mediator—the physical AMMR (Sec. 6.3). The verifier-tier comparison makes the same point economically: a weaker model escalates more, resolves less, and pays back part of its nominal price advantage in extra calls—while decisiveness itself is no guarantee of precision.

Future work follows the limitations surfaced by our own evaluation: extending the remnant filter to non-vertical structure artifacts, coverage-adaptive decomposition to resolve the mega-cluster paradox, close-range or re-rendered imagery for small low-reflectance objects, structural-change updating beyond the object level, automatic place segmentation to close the loop with the DK-SMF lineage, and embodied validation in which a robot plans against the graph—including its unverified residue—in the loop.

Supplementary Materials: The following supporting information can be downloaded at: <https://www.mdpi.com/article/10.3390/electronics1010000/s1>, Video S1: real-robot semantic mission execu-

tion — synchronized management console, Isaac Sim twin, and camera footage of the physical AMMR for the two filmed missions of Table 14 (goto_object tv; goto_place electrical_maintenance_area).

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Data Availability Statement: The TLS-SMF implementation is openly available at <https://github.com/kkk3449/tls-smf>; its supplementary/ directory additionally provides the owner-audited ground-truth sets for all five evaluated cluster sets (with grading provenance), the four-condition query benchmarks with per-claim scoring, the fixed pipeline configuration, and pointers to the exact prompt templates, forced answer schema, and synonym-tolerant scoring code used for every reported number. The processed semantic-object records and knowledge-graph snapshots are available from the corresponding author on reasonable request. The raw terrestrial laser scans (all three epochs of the test room, including the wall-removed T1 processing input, and the cafeteria stress scene; E57 format with SHA-256 checksums) are openly deposited on Zenodo under CC-BY 4.0 (DOI: <https://doi.org/10.5281/zenodo.22590752>). The determinism verification kit (version-pinned container recipe, reference checksums, and comparison script; Sec. 5.2) is included in the repository.

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Conflicts of Interest: Authors Sangmin Kim, Yonghyeon Song, and Byeongjun Kim are employed by CASELAB Co., Ltd.; author Tae-Yong Kuc serves as vice chairman of CASELAB Co., Ltd. The company provided access to the indoor facility in which the terrestrial laser scans were collected and, as the facility owner, confirmed the ground-truth object labels. The company had no role in the design of the study; in the collection, analyses, or interpretation of data; in the writing of the manuscript; or in the decision to publish the results. The remaining authors declare no conflicts of interest.

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